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Financially Constrained Arbitrage in Illiquid Markets

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Abstract

This paper derives arbitrage trading strategies taking into account the fact that the actions of arbitrageurs impact prices. This avoids the difficulty of having to rely on exogenous position limits to prevent infinite arbitrage profits. When arbitrageurs are financially constrained their trading strategies can be expressed as feedback functions of their capital, which in turn depends on the optimal amount traded. An important component of the trading by financially constrained arbitrageurs is done to guarantee future financial flexibility. It is this hedging component that explains why price deviations persist in spite of arbitrage. Financial constraints are also responsible for periods of excessively volatile prices and for the time variation in the correlation of price deviation across markets. The fact that the actions of regulated firms can influence the dynamics of prices on which minimum capital requirements are based raises important implications for the regulation of securities firms. (JEL G10, G20)

I. Introduction

In a letter to investors announcing the recasting of his Quantum Fund, George Soros wrote: “Quantum Fund is far too big and its activities too closely watched to be able to operate successfully in the market”.¹ A month earlier another hedge fund arbitrageur, Julian Robertson, echoed this same feeling when deciding to close Tiger Management which had become too big, undermining its ability to trade in and out of positions. Both these statements were made in a period of considerable market volatility and though pricing anomalies were frequent, according to both arbitrageurs, opportunities to exploit discrepancies were dwindling. Given the sheer size of some hedge fund portfolios it should not be surprising that their bets are often noticed by the market.²

Financial arbitrage is carried out with very large positions in assets taken by relatively few investors for short periods of time. Sophisticated arbitrageurs understand that their specialized trades often modify the path of prices. Some arbitrage strategies are done by “only four or five dealers”³, and in many markets “size extracts a price”. Although there is considerable evidence suggesting that market liquidity is often limited and that large trades move prices, the academic finance literature typically assumes that the optimal strategy for arbitrageurs can be derived ignoring the impact of arbitrage on prices. The few exceptions are the empirical studies of Mackinlay and Ramaswamy (1988) and Merrick (1989), which find evidence that path dependence in the basis of stock index futures contracts can be associated with the trading activities of arbitrageurs. Given that it is common practice for arbitrageurs when trading to consider the effects of marginal trading opportunities on the total value of their trades, it appears that competition between arbitrageurs can be viewed as less than atomistic. Deviations from the perfectly competitive case are not easily incorporated in a model of arbitrage trading because it raises the challenge of endogenizing the behavior of arbitrageurs given the impact of trades on market prices.⁴

In this paper we propose an analysis of arbitrage trading that attempts to capture some of these features in a simple and tractable way. Arbitrageurs are large and trade with noise traders, as well as with value-based traders. Asset prices deviate from the fundamental value and arbitrageurs aim to gain from buying cheap and selling expensive. However, since

¹See “High-flyers come back down to earth”, *The Financial Times*, April 29-30, 2000.

²Another interesting case is that of Pimco’s actions reportedly associated with “Bloody Thursday”, Feb 3, 2000, when the 30-year Treasury bond finished off a week with a steep rise of more than 2%, while mortgage bonds plummeted. According to the *Wall Street Journal*, Pimco bought large amounts of the 30-year Treasury bond and dumped mortgage bonds and triggered panicked trading. In Mr. William Gross, Pimco’s manager’s words, “I think we were the spark to the market tumult. Others found out about the trades too quickly, and we couldn’t do enough without driving the long bond to ridiculous levels.” *The Wall Street Journal*, March 2000.

³See Lowenstein (2000) page 176. The presence of a few dominant traders is more common than usually thought. One case that drew considerable visibility is Long Term Capital Management. Another is Metallgesellschaft’s US subsidiary (MG Corp). MG’s NYMEX positions in oil futures contracts were so large that they became the major influence on the market and their trades were common knowledge. See Mello and Parsons (1995).

⁴This point was first made in Brennan and Schwartz (1990).

arbitrageurs know that their actions affect market prices, it is necessary to depart from the assumption that the impact of arbitrage on price is known before the arbitrage strategy is derived.

A realistic model of arbitrage must also take into account that arbitrage traders back their trading activities with own capital. Examples include the capital adequacy directives establishing minimum capital requirements for the trading books of banks and securities firms, the system of margins imposed by clearing brokers, as well as margins on futures contracts and on leveraged equity accounts. Financial constraints can make a significant difference to the optimal level of arbitrage trading, as well as to the time series properties of market prices. In the presence of financial constraints based on market values, trading strategies can be expressed as feedback functions of the arbitrageur's current capital, which in turn relates to the amount an arbitrageur chooses to trade. As a result, a model of arbitrage trading must consider the costs imposed by the leverage constraints endogenously.⁵

In addition to paying careful consideration to how marginal trades impact on the total position, financially constrained arbitrageurs try to avoid violations of the capital constraints caused by adverse shocks to prices. Reducing the likelihood of violating the capital balances effectively means biasing arbitrage positions towards zero, despite the existence of profit opportunities. This is because a zero position in the asset is immune to adverse shocks which affect the value of the portfolio. Although the bias towards the zero position has opportunity costs associated with forgone profits, it has the benefit of reducing the likelihood of default and of increasing financial flexibility. Interestingly, the hedging motive makes financially constrained arbitrageurs at times trade larger quantities than unconstrained arbitrageurs faced with similar opportunities.

In comparing the behavior of prices, we find that when arbitrageurs are financially unconstrained, the expected future asset price is always equal to its expected fair value. However, with financially constrained arbitrageurs the market price tends to deviate from its fundamental value for successive periods. The reason is that in his desire to gain financial flexibility, a long arbitrageur sells more of the security when noise traders want to buy it than he buys of the security when noise traders want to sell it. Therefore, the mean value of the trade of a constrained arbitrageur with a long position is negative. This has the effect of driving the price below the fundamental value. The opposite occurs when the arbitrageur is initially short in the asset. Persistent price deviations occur and are then the result of arbitrageurs being financially constrained, since no price deviations are expected if arbitrageurs are unconstrained.⁶

Although it is not our purpose to explain financial crisis, in our model financial constraints cause observed anomalies, such as excessively volatile prices, and regime switching volatility, with calm periods turning into turbulent periods. Contrary to the case of a market with unconstrained arbitrageurs, where volatility is constant, in a market with financially constrained arbitrageurs price volatility depends on the positions held by arbitrageurs. Nor-

⁵Several other articles have also addressed the effects of financial constraints on the trading strategies of rational traders - see, for example, Grossman and Vila (1992). However, in these models trading strategies do not modify the price process and leverage constraints are evaluated in the presence of exogenously imposed leverage costs.

⁶Long run price inefficiencies have also been explained by transaction costs, see Tuckman and Vila (1992), and short horizons, see Dow and Gordon (1994).

mally, arbitrageurs reduce price volatility and improve market liquidity. However, there are periods of especially volatile prices, because arbitrageurs, after an adverse shock, become financially constrained and are forced to liquidate positions. Arbitrage firms with less capital as a result of losses may find themselves in violation of their own guidelines regarding the proper ratio of capital to assets. Thus, they sell more. Instead of contributing to reverse the effect of the liquidity shock, during these periods arbitrageurs cause the destabilization of market prices. As traders put it: “as people liquidate, volatility moves up”.⁷ Higher price volatility also occurs when noise trading increases and the elasticity of demand for trade of valued-based investors declines.⁸

With shifting patterns in volatility, regulatory constraints on arbitrage trading, such as margin calls and tighter capital requirements, need to be examined very carefully. We find, for example, that when financially fragile arbitrageurs find it impossible to raise additional capital from reluctant investors such constraints can leave markets in a funk. Also, less than perfect rules, such as the standard Value at Risk (VaR) measure, can contribute to the perverse regulation of trading risk and cause inefficiencies in the financial system. We assert that VaR rules that ignore that trading by financial firms have a direct impact on prices do not accurately measure the trading risk of these firms’ portfolios and therefore lead to distortions.

The actions of financially constrained arbitrageurs are also related to the correlations of prices across markets. In a multi-asset environment we show that correlations between markets increase significantly in periods of instability, a fact that has been well documented in recent empirical studies. Shocks that are unique to one market can affect the financial condition of arbitrageurs with positions in that market and force them to liquidate holdings in other assets, which contributes to the propagation of volatility across markets. This idea of contagion is related to that found in Kodres and Prisker (1998), where informed investors reduce asset positions in various assets during market declines to meet portfolio balancing needs. Undiversified investors in unrelated markets are not clear about the motives of informed investors and sell as well. The final result is that a negative shock in an asset spreads to other assets even when nothing fundamentally changed. In our model these effects are even more pronounced because the actions by arbitrageurs impact both the level and the volatility of market prices.⁹

Some of the points made in our paper are presented in Shleifer and Vishny (1997). Their paper considers an arbitrageur working for investors who observe the returns but not the investment opportunities available. Their conclusions are similar to some of those presented here, namely that the fully invested arbitrageur may need to liquidate positions after a negative shock that results in immediate losses caused from an increase in the mispricing,

⁷This paper is not the first to point out this anomaly. Gennotte and Leland (1990) show that rational traders facing portfolio insurance constraints put downward pressure on prices and amplify volatility in declining markets.

⁸Yuan (1999) attempts to explain financial crisis arising from informational differences between rational speculators facing borrowing constraints and liquidity traders with upward sloping demand functions. When the asset price is high liquidity traders absorb more of the random asset supply. When the asset price is low liquidity traders are increasingly resistant to absorb any supply because they do not know whether arbitrage trading is informational driven or because of borrowing constraints.

⁹See also Xiong, W. and A. Kyle “Contagion as a Wealth Effect”, forthcoming in the Journal of Finance.

mispricing that improves the future returns to the position. This will happen when the arbitrageur is unable to raise the additional capital. The authors argue, without an explicit model, that risk averse arbitrageurs may liquidate positions even when they are not fully invested.

More recent work, such as Liu and Longstaff (2000) and Loewenstein and Willard (2000), also point out the risks, as we do here, of arbitrage trading with credit constraints. Liu and Longstaff solve for the trading policy of risk-averse arbitrageurs faced with margin calls when security prices converge at some known time to zero. They show that arbitrageurs trade conservatively, a result that resembles our finding that arbitrageurs avoid violating the capital requirements by underinvesting in arbitrage opportunities. Loewenstein and Willard avoid the feature of exogenous prices in Liu and Longstaff and show that mispricing persists despite arbitrage activity. The complementarity between our work with large risk neutral arbitrageurs and that of these authors, which rely on competitive risk averse traders, gives an indication of the robustness of the results. Despite important economic differences, the papers reach the same conclusions, that arbitrage does not yield infinite profits and may involve considerable risks. Finally, Xiong (2001) emphasizes the liquidity crisis created by the arbitrageurs limited capacity to bear risks. When losses are significant, arbitrageurs unload their positions and this contributes to the destabilization of markets, since arbitrageurs trade in the same direction as noise traders. His paper in conjunction with ours, Loewenstein and Willard and the literature in market microstructure shows that the notion of market liquidity relates as much to the risks present in the economy as it does to the type and role played by different traders and institutions.

The remainder of the paper is organized as follows. Section II describes the economy and the optimal trading strategy of a monopolist unconstrained arbitrageur. The section also contains our main results on financially constrained arbitrage strategies and presents the moments of the distribution of price deviations. Section III extends the results to a market with multiple arbitrageurs competing with each other. Section IV considers the case of arbitrage trading in multiple risky securities and derives implications for price behavior across markets. Section V discusses policy issues related to the regulation of arbitrage activity. Section VI concludes.

II. The Model

This section presents a dynamic strategy for a risk-neutral arbitrage trader who trades in a market for a security with a population of value-based traders and noise traders. To facilitate the exposition, we start with no arbitrage traders. The fundamental value of the security is not known when trade occurs. The dividend paid by the security, d_t , evolves as

$$d_t = d_{t-1} + \sigma_d \varepsilon_t \quad (1)$$

where, σ_d is the per period volatility and ε_t is a zero mean, $E[\varepsilon_t] = 0$, unit variance, $E[\varepsilon_t^2] = 1$, innovation in the dividend at time t .¹⁰ Risk-neutral traders value the asset as

¹⁰At this point we do not need additional assumptions regarding the distribution of ε_t . In later sections we assume that ε_t has a symmetric distribution and in section V we assume that $\varepsilon_t \sim N(0,1)$. But these

the discounted expected stream of future dividends:

$$I_t = E_t \left[\sum_{n=0}^{\infty} \delta^n d_{t+n} \right] = d_t \left[\sum_{n=0}^{\infty} \delta^n \right] = \frac{d_t}{1-\delta} = \frac{d_{t-1}}{1-\delta} + \frac{\sigma_d}{1-\delta} \varepsilon_t = I_{t-1} + \sigma_I \varepsilon_t \quad (2)$$

where, I_t is the intrinsic value of the asset prior to the payment of dividend at time t , $\sigma_I = \frac{\sigma_d}{1-\delta}$ and $\delta < 1$ is the discount factor.

The fair value is never observed directly but is revealed by the dividend paid by the security. The market price of the security is denoted by P_t . We assume that there are two components to liquidity trades in the market for the security. The first is a pure noise component, $\sigma_F \varepsilon_t$. The other component represents the discretionary actions of value based traders, who respond to the deviation of the market price from its fair price. In particular, a value based trader contemplating a purchase of the asset will do so if $P_t < I_{t-1}$, since this means that the asset looks “cheap”.¹¹ We therefore assume that the total demand by both noise and value-based traders is given by

$$D_t^L = -\beta (P_t - I_{t-1}) + \sigma_F \varepsilon_t \quad (3)$$

where D_t^L is the flow of liquidity trades in the interval $(t-1, t)$, and $\beta > 0$. Notice that the first component of the demand is the price sensitive demand and the second is a flow of random liquidity orders from those who trade and are not price sensitive. For simplicity, we assume that same shock affects both the asset value and the liquidity demand.¹²

The price of the security is set by market clearing. In the absence of arbitrage traders this price is determined by setting D_t^L to zero, giving $P_t = I_{t-1} + \frac{\sigma_F}{\beta} \varepsilon_t$. The value-based traders gain from trading the security if they are able to purchase it at a price less than value, that is if $P_t < I_t$, and sell it at a price greater than value, if $P_t > I_t$. This is true if $\frac{\sigma_F}{\beta} > \sigma_I$, which holds if $\beta < \frac{\sigma_F}{\sigma_I}$; $\beta > \frac{\sigma_F}{\sigma_I}$ is unlikely to be observed since the value based traders lose money from trading in this case and have no incentive to trade.

A. Trading Strategy of an Unconstrained Monopolist Arbitrageur

The price of the security in the absence of arbitrageurs is different from its value. The magnitude of the difference is $\left| \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \right|$. The difference results from the fact that information related to the value of the security is incorporated into prices via the demand of liquidity traders. Arbitrageurs, will attempt, in light of their knowledge of the process generating liquidity trades to profit from these price deviations. The sequence of events in each period is as follows: (i) the liquidity shock is observed, (ii) the security is traded, (iii) the security pays a dividend and (iv) the arbitrageurs consume the profits.

Let, θ_{t-1} , represent the number of units of the security held by the arbitrageur after trading at date $t-1$ (prior to trading at date t); the quantity that he trades at date t is

additional assumptions are made primarily to simplify the presentation.

¹¹The assumption here is that value based traders submit demand schedules, that is, they specify the quantity that they will trade as a function of price. They are willing to purchase the asset when $P_t < I_{t-1}$ and sell it when $P_t > I_{t-1}$ since $E_{t-1}[I_t] = I_{t-1}$.

¹²This assumption can be relaxed very easily - two correlated shocks can be used, where the shocks have a correlation ρ . This makes the notation complicated and does not add to the intuition of the problem.

$\theta_t - \theta_{t-1}$. Given that the demand for the security from value-based traders is not perfectly elastic, trading by the arbitrageur affects the market price of the security. If, as before, the liquidity trader demand for the security is given by D_t^L , the market price is set by the market clearing condition that $D_t^L + \theta_t - \theta_{t-1} = 0$. This gives the following expression for the price of the security:

$$P_t = I_{t-1} + \frac{\sigma_F}{\beta} \varepsilon_t + \frac{\theta_t - \theta_{t-1}}{\beta} \quad (4)$$

The expression above describes how the demand flow from arbitrageurs affects the market price of the asset. If the arbitrageur is buying the security ($\theta_t - \theta_{t-1} > 0$) the price, P_t , will rise relative to its fundamental value. The reverse will be true if the arbitrageur is selling the security ($\theta_t - \theta_{t-1} < 0$).

The arbitrageur's profit from trading the security is given as:

$$\text{Trading Profit}(\theta_t - \theta_{t-1}) = (\theta_t - \theta_{t-1}) (I_t - P_t) \quad (5)$$

where, the gain from buying (selling) the security at a price that is lower (higher) than its value is sensitive to the choice of quantity traded. The total profit earned by the arbitrageur from holding a position in the security is the sum of the trading profit and the gain or loss on the existing position. The gain on the existing position is the sum of the change in value and the cost of financing the position net of dividend.¹³ The total profit is given as

$$\begin{aligned} \text{Total Profit}(\theta_t) &= \text{Trading Profit}(\theta_t - \theta_{t-1}) + \text{Gain/Loss on Existing Position} \\ &= \text{Trading Profit}(\theta_t - \theta_{t-1}) + \left(\begin{array}{l} \text{Change in Value of Existing Position} \\ - \text{Financing Cost Net of Dividend} \end{array} \right) \\ &= (\theta_t - \theta_{t-1}) (I_t - P_t) + \left(\theta_t (I_{t+1} - I_t) - \left(\theta_t (I_t - d_t) \frac{1}{\delta} - \theta_t I_t \right) \right) \\ &= (\theta_t - \theta_{t-1}) (I_t - P_t) + \theta_t (I_{t+1} - I_t) \end{aligned} \quad (6)$$

where the financing cost is the interest payments less the dividend received. The expected value of the above total profit based on information available at the time of trading is equal to the trading profit.

If the arbitrageur consumes the trading profit each period, the arbitrageur's objective function of maximizing the discounted expected utility of (infinite horizon) consumption starting period t , is given as¹⁴

¹³The financing cost represents the actual interest payments that have to be made if the position is financed using borrowed funds. If the position is financed using the arbitrageurs own funds, the financing cost represents the opportunity cost of the funds.

¹⁴We compute the arbitrageur's expected utility based on the consumption of the trading profit because this is the marginal amount generated as a result of trading. The remaining terms in the expression for total profit are the same as what would have been generated by a simple buy and hold portfolio strategy.

$$\begin{aligned}
V_U(\theta_{t-1}, \varepsilon_t) &= \max_{\{\theta_{t+n}\}_{n=0, \dots, \infty}} \sum_{n=0}^{\infty} \delta^n E_t [\text{Consumption}(t+n)] \\
&= \max_{\{\theta_{t+n}\}_{n=0, \dots, \infty}} \sum_{n=0}^{\infty} \delta^n E_t [\text{Trading Profit}(\theta_{t+n})] \\
&= \max_{\theta_t} E_t [(\theta_t - \theta_{t-1})(I_t - P_t) + \delta V_U(\theta_t, \varepsilon_{t+1})] \\
&= \max_{\theta_t} (\theta_t - \theta_{t-1})(I_t - P_t) + \delta E_t [V_U(\theta_t, \varepsilon_{t+1})]
\end{aligned} \tag{7}$$

The solution to the optimization problem is $\theta_{t+n}^{U*} = \theta_{t+n-1} - \frac{\beta}{2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_{t+n}$. The security price over adjusts and the arbitrageur buys the security, that is $\theta_{t+n}^{U*} > \theta_{t+n-1}$, when noise traders are net sellers, $\varepsilon_{t+n} < 0$; and the arbitrageur sells the security, that is $\theta_{t+n}^{U*} < \theta_{t+n-1}$, when noise traders are net buyers, $\varepsilon_{t+n} > 0$. Both value-based traders and the arbitrageur profit from trading against the noise traders.

Note that in this simple setting the quantity traded is not a function of the current position or of any other variable. Substituting the optimally traded quantity into the expression for the price of the security, gives $P_t = I_{t-1} + \frac{1}{2} \left(\frac{\sigma_F}{\beta} + \sigma_I \right) \varepsilon_t$, and into the objective function of the monopolist arbitrageur gives the maximized profit as

$$V_U(\theta_{t-1}, \varepsilon_t) = \frac{\beta}{4} \left(\frac{\sigma_F}{\beta} - \sigma_I \right)^2 \left[\varepsilon_t^2 + \left(\frac{\delta}{1-\delta} \right) \right] \tag{8}$$

Arbitrage traders in financial markets are often required to back their trading positions with own capital. Examples include banks subject by regulators minimum capital requirements and margin requirements in futures contracts. Let us next analyze the problem of an arbitrageur who faces such capital requirements.

B. Trading Strategy of a Financially Constrained Monopolist Arbitrageur

The most common constraints faced by traders in financial markets are minimum capital constraints and margin requirements. In this sub-section we consider the case of an arbitrageur who faces a minimum capital requirement in the form of a bankruptcy constraint; discussion of other types of constraints is delayed until a later section.¹⁵ The sequence of events in each period is as follows: (i) the liquidity shock is observed, (ii) a determination of the arbitrageur's solvency is made, (iii) the security is traded with a liquidation of the arbitrageur's portfolio if he is insolvent, (iv) the security pays a dividend and (v) the arbitrageur consumes his profits. Figure 1 shows the sequence of events.

In order to understand the implications of the constraints, we start by modelling the evolution of the arbitrageur's capital balance. Define three types of capital: the Economic

¹⁵A very simple form of the bankruptcy constraint is considered here - the arbitrageur is forced to leave the market immediately liquidating his position if one measure of his capital becomes zero. More complicated forms of the constraint would include different liquidation strategies and different liquidation triggers. Almgren and Chriss (1998) show that a liquidation strategy of the type considered here is the strategy that has the minimum variance of liquidation proceeds.

Capital,

$$EC_t = \theta_{t-1} I_t - B_t \quad (9)$$

is defined in terms of the intrinsic value of the asset, the Market Capital is defined in terms of the market price of the asset

$$MC_t = \theta_{t-1} P_t - B_t, \quad (10)$$

while the regulatory capital

$$RC_t = \theta_{t-1} P_t (\theta_t = 0, \varepsilon_t) - B_t \quad (11)$$

is defined in terms of the liquidation value of the asset, $P_t (\theta_t = 0, \varepsilon_t)$. In all cases B_t is the principal amount owed to creditors at the end of period t . Using the above definitions we get $EC_t = RC_t + \theta_{t-1} (I_t - P_t (\theta_t = 0, \varepsilon_t))$.

If consumption in each period is equal to the trading profit generated in that period the change in economic capital from time $t - 1$ to time t is given by ¹⁶

$$EC_t = EC_{t-1} + \theta_{t-1} (I_t - I_{t-1}). \quad (12)$$

Consumption refers to the incremental amount available after paying a return on the economic capital. This incremental amount is $(\theta_t - \theta_{t-1}) (I_t - P_t)$, while the total amount has the additional term $(\frac{1}{\delta} - 1) EC_{t-1}$, the interest paid on the arbitrageur's economic capital. The evolution of debt, using the above, is given as

$$B_t = B_{t-1} + (\theta_{t-1} - \theta_{t-2}) I_{t-1}, \quad (13)$$

the evolution of the market capital as

$$MC_t = MC_{t-1} + \theta_{t-1} (P_t - P_{t-1}) - (\theta_{t-1} - \theta_{t-2}) (I_{t-1} - P_{t-1}). \quad (14)$$

and the evolution of the regulatory capital as

$$RC_t = RC_{t-1} + \theta_{t-2} (I_{t-1} - P_{t-1} (\theta_{t-1} = 0, \varepsilon_{t-1})) - \theta_{t-1} (I_t - P_t (\theta_t = 0, \varepsilon_t)) + \theta_{t-1} (I_t - I_{t-1}) \quad (15)$$

Ultimately, it is the amount of capital remaining on liquidation, RC_t , that must satisfy some exogenously imposed reservation value. The arbitrageur can continue trading the security as long as this measure of regulatory capital is greater than zero (although any other cut-off would work) and that he is forced to liquidate all positions and leave the market if the

¹⁶The assumption here is that the capital is computed prior to consumption. If the arbitrage trader passes a solvency check and there are current losses from trading, new equity is raised to fund these losses. This inflow represents current negative consumption. If the arbitrage trader does not pass the solvency test, liquidation follows. If the arbitrageur were allowed to raise capital between the realization of the liquidity shock and the solvency test, he would be able to raise new capital, ruling out bankruptcy even if it were optimal to liquidate. Schleifer and Vishny (1997) explain why it is likely to be difficult for an arbitrageur to raise new capital at short notice.

above measure of capital is non-positive. This allows us to write the constrained monopolist arbitrageur's optimization problem as

$$V_C(RC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \underset{\theta_t}{Max} (\theta_t - \theta_{t-1})(I_t - P_t) + \delta E[V_C(RC_t, \theta_t, \varepsilon_{t+1})] & RC_t > 0 \\ V_L(\theta_{t-1}, \varepsilon_t) & RC_t \leq 0 \end{cases} .$$

$$= \begin{cases} \underset{\theta_t}{Max} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} - (\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t + \delta E[V_C(RC_t, \theta_t, \varepsilon_{t+1})] & RC_t > 0 \\ \theta_{t-1}\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & RC_t \leq 0 \end{cases} \quad (16)$$

where, $V_L(\theta_{t-1}, \varepsilon_t) = \theta_{t-1}\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t - \frac{\theta_{t-1}^2}{\beta}$, the cost of liquidating the arbitrageur's position is obtained by setting θ_t to zero.

Using the relation between economic and regulatory capital, $EC_t = RC_t + \theta_{t-1}(I_t - P_t(\theta_t = 0, \varepsilon_t))$ and the evolution of economic capital, $EC_t = EC_{t-1} + \theta_{t-1}(I_t - I_{t-1})$, the condition $RC_t > 0$ can now be replaced by an equivalent condition $EC_{t-1} > \theta_{t-1}\left(\frac{\theta_{t-1}}{\beta} - \frac{\sigma_F}{\beta}\varepsilon_t\right)$ which after algebraic manipulation gives $\frac{\beta}{\sigma_F}EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t$. The right hand side of the inequality is the size of the loss-causing shock, and the condition indicates the maximum magnitude of an adverse shock that the arbitrageur can bear without being forced to liquidate. The arbitrageur's optimization problem is now simplified to:

$$U_C(EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \underset{\theta_t}{Max} \left\{ \begin{aligned} & -\frac{(\theta_t - \theta_{t-1})^2}{\beta} \\ & -(\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ & + \delta E[U_C(EC_t, \theta_t, \varepsilon_{t+1})] \end{aligned} \right\} & \frac{\beta}{\sigma_F}EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t \\ \theta_{t-1}\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \frac{\beta}{\sigma_F}EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} \leq -\theta_{t-1}\varepsilon_t \end{cases} \quad (17)$$

where, $U_C(EC_{t-1}, \theta_{t-1}, \varepsilon_t)$ is equivalent to $V_C(RC_{t-1}, \theta_{t-1}, \varepsilon_t)$.

Let $f(EC, \theta, \varepsilon)$ be the optimized value of the objective function in the continuation region, that is $f(EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \underset{\theta_t}{Max} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} - (\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t + \delta E[U_C(EC_t, \theta_t, \varepsilon_{t+1})]$

when $\frac{\beta}{\sigma_F}EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t$. For $E[U_C(EC_t, \theta_t, \varepsilon_{t+1})]$ we can write

$$E[U_C(EC_t, \theta_t, \varepsilon_{t+1})] = E \left[\begin{cases} f(EC_t, \theta_t, \varepsilon_{t+1}) & \frac{\beta}{\sigma_F}EC_t - \frac{\theta_t^2}{\sigma_F} > -\theta_t\varepsilon_{t+1} \\ \theta_t\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_{t+1} - \frac{\theta_t^2}{\beta} & \frac{\beta}{\sigma_F}EC_t - \frac{\theta_t^2}{\sigma_F} \leq -\theta_t\varepsilon_{t+1} \end{cases} \right] \quad (18)$$

The expectation in the above is over ε_{t+1} . Let $L_{t+1} \equiv -\frac{\beta}{\sigma_F}\frac{EC_t}{\theta_t} + \frac{\theta_t}{\sigma_F}$, for $\theta_t > 0$ ($\theta_t < 0$) we can rewrite $\frac{\beta}{\sigma_F}EC_t - \frac{\theta_t^2}{\sigma_F} > -\theta_t\varepsilon_{t+1}$ as $\varepsilon_{t+1} > L_{t+1}$ ($L_{t+1} > \varepsilon_{t+1}$). Using this we can write for $\theta_t > 0$ ¹⁷

$$E[U_C(EC_t, \theta_t, \varepsilon_{t+1})] = \int_{-\infty}^{L_{t+1}} \left(\theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_{t+1} - \frac{\theta_t^2}{\beta} \right) g(\varepsilon_{t+1}) d\varepsilon_{t+1} + \int_{L_{t+1}}^{\infty} f(EC_t, \theta_t, \varepsilon_{t+1}) g(\varepsilon_{t+1}) d\varepsilon_{t+1} \quad (19)$$

¹⁷For $\theta_t < 0$ the corresponding expression is

$$E[V_C(X_t, \theta_t, \varepsilon_{t+1})] = \int_{L_{t+1}}^{\infty} \left(\theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_{t+1} - \frac{\theta_t^2}{\beta} \right) g(\varepsilon_{t+1}) d\varepsilon_{t+1} + \int_{-\infty}^{L_{t+1}} f(X_t, \theta_t, \varepsilon_{t+1}) g(\varepsilon_{t+1}) d\varepsilon_{t+1}$$

where $g(\varepsilon_{t+1})$ is the probability density function associated with ε_{t+1} . This gives

$$E[U_C(EC_t, \theta_t, \varepsilon_{t+1})] = -\theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) g(L_{t+1}) - \frac{\theta_t^2}{\beta} G(L_{t+1}) + \int_{L_{t+1}}^{\infty} f(EC_t, \theta_t, \varepsilon_{t+1}) g(\varepsilon_{t+1}) d\varepsilon_{t+1} \quad (20)$$

where $G(\cdot)$ is the cumulative density function associated with ε_{t+1} . Substituting back into the arbitrageur's optimization problem gives

$$U_C(EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \max_{\theta_t} \left\{ \begin{aligned} & -\frac{(\theta_t - \theta_{t-1})^2}{\beta} - (\theta_t - \theta_{t-1}) \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \\ & + \delta \int_{L_{t+1}}^{\infty} f(EC_t, \theta_t, \varepsilon_{t+1}) g(\varepsilon_{t+1}) d\varepsilon_{t+1} \\ & - \delta \theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) g(L_{t+1}) - \delta \frac{\theta_t^2}{\beta} G(L_{t+1}) \end{aligned} \right\} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1} \varepsilon_t \\ \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} \leq -\theta_{t-1} \varepsilon_t \end{cases} \quad (21)$$

The functional form of $f(EC, \theta, \varepsilon)$ is not known. However, from the above we have the following relationship that can be used to solve for $f(EC, \theta, \varepsilon)$

$$f(EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \max_{\theta_t} \begin{aligned} & -\frac{(\theta_t - \theta_{t-1})^2}{\beta} - (\theta_t - \theta_{t-1}) \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \\ & + \delta \int_{L_{t+1}}^{\infty} f(EC_t, \theta_t, \varepsilon_{t+1}) g(\varepsilon_{t+1}) d\varepsilon_{t+1} \\ & - \delta \left(\theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) g(L_{t+1}) + \frac{\theta_t^2}{\beta} G(L_{t+1}) \right) \end{aligned} \quad (22)$$

given that EC_{t-1}, θ_{t-1} and ε_t satisfy $\frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} + \theta_{t-1} \varepsilon_t > 0$. The first order condition is

$$0 = -\frac{2(\theta_t - \theta_{t-1})}{\beta} - \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \delta \left(\left(\frac{\sigma_F}{\beta} - \sigma_I \right) g(L_{t+1}) + \frac{2\theta_t}{\beta} G(L_{t+1}) \right) - \delta f(EC_t, \theta_t, L_{t+1}) g(L_{t+1}) \frac{\partial L_{t+1}}{\partial \theta_t} + \delta \int_{L_{t+1}}^{\infty} \frac{\partial}{\partial \theta_t} f(EC_t, \theta_t, \varepsilon_{t+1}) g(\varepsilon_{t+1}) d\varepsilon_{t+1} - \delta \left(\theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \frac{\partial}{\partial \theta_t} g(L_{t+1}) + \frac{\theta_t^2}{\beta} g(L_{t+1}) \frac{\partial L_{t+1}}{\partial \theta_t} \right) \quad (23)$$

The above problem does not have an analytical solution. We proceed by first solving the problem numerically. Then we provide an analytical approximation which matches the properties of the numerical solution. This closed-form approximation has the advantage that it allows us to take the analysis much further and provide additional intuition describing equilibrium price processes and trading strategies not possible using the numerical solution.

1. Numerical Solution of the Problem

The difficulty in directly solving the above expression analytically is that we do not know the functional form of $f(EC, \theta, \varepsilon)$. What we do have is an expression that relates the value of $f(\cdot, \cdot, \cdot)$, evaluated at the current levels of the state variables, to its expected value evaluated over the level of the state variables and the optimized control variable next period. One way to numerically solve for $f(\cdot, \cdot, \cdot)$ is to start with some “terminal” values at a set of grid points and iterate back till $f(\cdot, \cdot, \cdot)$ converges. The problem with doing this directly on $f(\cdot, \cdot, \cdot)$ is that with three state variables a dense grid makes each iteration time consuming,

while a sparse grid results in oscillatory or divergent behavior. For this reason we take a step back to the original problem described in (17) and repeated here for convenience

$$U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \text{Max}_{\theta_t} \left\{ \begin{array}{l} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} \\ -(\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ +\delta E[U_C (EC_t, \theta_t, \varepsilon_{t+1})] \end{array} \right\} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t \\ \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} \leq -\theta_{t-1}\varepsilon_t \end{cases} \quad (24)$$

and define $J (EC_{t-1}, \theta_{t-1}) = E [U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t)]$. Using this, the fact that $EC_t = EC_{t-1} + \theta_{t-1}\sigma_I\varepsilon_t$ and taking expectations of both sides of (24) we can write for $\theta_{t-1} > 0$ ¹⁸

$$J (EC_{t-1}, \theta_{t-1}) = \int_{-\infty}^{L_t} \left(\theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} \right) g(\varepsilon_t) d\varepsilon_t + \int_{L_t}^{\infty} \text{Max}_{\theta_t} \left\{ \begin{array}{l} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} - (\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ +\delta J (EC_{t-1} + \theta_{t-1}\sigma_I\varepsilon_t, \theta_t) \end{array} \right\} g(\varepsilon_t) d\varepsilon_t \quad (25)$$

where, as before, $L_t = \frac{\theta_{t-1}}{\sigma_F} - \frac{\beta}{\sigma_F} \frac{EC_{t-1}}{\theta_{t-1}}$ and $g(\varepsilon_t)$ is the probability density function associated with ε_t . In this case we reduce the problem to two state variables and can use a sufficiently dense grid that results in fast convergence.

Once we have $J (EC_{t-1}, \theta_{t-1})$ we can solve for $U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t)$ by substituting back into the arbitrageurs optimization problem. This gives

$$U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \text{Max}_{\theta_t} \left\{ \begin{array}{l} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} \\ -(\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ +\delta J (EC_{t-1} + \theta_{t-1}\sigma_I\varepsilon_t, \theta_t) \end{array} \right\} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t \\ \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} \leq -\theta_{t-1}\varepsilon_t \end{cases} \quad (26)$$

It is now possible to solve for the optimal trading strategy and to characterize $U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t)$. In the next sub-section we use the information obtained from the numerical solution to build an analytically tractable approximation to the arbitrageur's problem. While we can get all the results reported in the paper using the numerical solution, the analytical approximation allows us to gain greater intuitive insights into the problem.

¹⁸For $\theta_{t-1} < 0$ the expression is

$$J (X_{t-1}, \theta_{t-1}) = \int_{L_t}^{\infty} \left(\theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} \right) g(\varepsilon_t) d\varepsilon_t + \int_{-\infty}^{L_t} \left(\text{Max}_{\theta_t} \left\{ \begin{array}{l} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} - (\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ +\delta J (X_{t-1} + \theta_{t-1}\sigma_I\varepsilon_t, \theta_t) \end{array} \right\} \right) g(\varepsilon_t) d\varepsilon_t$$

2. An Approximate Analytical Solution of the Problem

Consider the arbitrageur's problem repeated below for convenience

$$U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \begin{matrix} \text{Max}_{\theta_t} \left\{ \begin{matrix} -\frac{(\theta_t - \theta_{t-1})^2}{\beta} \\ -(\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ +\delta E[U_C (EC_t, \theta_t, \varepsilon_{t+1})] \end{matrix} \right\} \end{matrix} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t \\ \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} \leq -\theta_{t-1}\varepsilon_t \end{cases} \quad (27)$$

The approximation replaces the optimized value of the future from period $t + 1$ onwards, $E[U_C (EC_t, \theta_t, \varepsilon_{t+1})]$, with a particular separable functional form. The approximation is based on the following observation: The optimized value of $U_C (EC_t, \theta_t, \varepsilon_{t+1})$ is known at the two boundaries described by a financially unconstrained, $EC_t = \infty$ ($RC_{t+1} = \infty$), and a zero capital, $EC_t \leq \frac{\theta_t^2}{\beta} - \frac{\sigma_F}{\beta} \theta_t \varepsilon_{t+1}$ ($RC_{t+1} \leq 0$), arbitrageur. In the first case the optimized value function is given by $V_U (\theta_t, \varepsilon_{t+1})$ (as in equation (8)). In the second case, the arbitrageur is forced to liquidate his position and not trade ever again, with a value function that is given by $V_L (\theta_t, \varepsilon_{t+1})$. The value of the optimal trading strategy of a financially constrained arbitrageur with positive regulatory capital $\frac{\theta_t^2}{\beta} - \frac{\sigma_F}{\beta} \theta_t \varepsilon_{t+1} < EC_t < \infty$ ($0 < RC_{t+1} < \infty$), must lie between V_L and V_U . From the numerical solution we know that the value of arbitrage trading is a concave function of the amount of capital balances available. We use this fact to approximate the $U_C (EC_t, \theta_t, \varepsilon_{t+1})$ term in the above optimization problem by a weighted sum of the value function at the two boundaries, V_U and V_L . The problem is then written as:

$$U_C (EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \begin{matrix} \text{Max}_{\theta_t} \left\{ \begin{matrix} -(\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t - \frac{(\theta_t - \theta_{t-1})^2}{\beta} \\ +\delta E \left[\begin{matrix} (1 - \pi(EC_t, \varepsilon_{t+1})) V_U (\theta_t) \\ +\pi(EC_t, \varepsilon_{t+1}) V_L (\theta_t) \end{matrix} \right] \end{matrix} \right\} \end{matrix} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} > -\theta_{t-1}\varepsilon_t \\ \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \frac{\beta}{\sigma_F} EC_{t-1} - \frac{\theta_{t-1}^2}{\sigma_F} \leq -\theta_{t-1}\varepsilon_t \end{cases} \quad (28)$$

where the weights, $\pi(EC_t, \varepsilon_{t+1})$ and $1 - \pi(EC_t, \varepsilon_{t+1})$, given to each component of the value function next period, V_L and V_U , are function of the arbitrageur's capital balances next period.

A consistent choice of π 's would be one which equals one if $EC_t \leq \frac{\theta_t^2}{\beta} - \frac{\sigma_F}{\beta} \theta_t \varepsilon_{t+1}$, that is when the trader has zero regulatory capital balance and zero if the trader has infinite required capital available. The π 's should be monotonically decreasing functions of the capital balance available to the trader, since a higher level of capital should imply a lower likelihood of forced liquidation. Also, the π 's should be such that the approximation above should be a concave function of available capital.

The arbitrageur's capital balance next period is a function of his current capital balance, the existing position in the asset and the realization of the liquidity shock. A higher level of existing capital balance, EC_{t-1} , translates into a higher future capital balance and, therefore, should translate into lower π 's. If the arbitrageur has an initial long position, a negative liquidity shock decreases his capital and should be reflected in a higher π since for $\theta_t > 0$ EC_t is increasing in the size of the shock; for $\theta_t < 0$ EC_t is decreasing in the size of the

shock resulting in higher π' s for positive shocks.

A functional form that satisfies the above requirements for the π' s is

$$\pi(EC_t, \varepsilon_{t+1}) = \frac{a}{a + |\bar{\theta}_t(EC_t, \varepsilon_{t+1})|} \quad (29)$$

where, a is a constant that depend on the value of the parameters σ_F , σ_I and β and $\bar{\theta}_t(EC_t, \varepsilon_{t+1})$ is the maximum position that the trader can hold in the risky asset without being forced to liquidate at $t + 1$; that is, $\bar{\theta}_t(EC_t, \varepsilon_{t+1})$ solves

$$0 = \frac{\bar{\theta}_t^2}{\beta} - \frac{\sigma_F}{\beta} \bar{\theta}_t \varepsilon_{t+1} - EC_t. \quad (30)$$

Of the two roots, $\bar{\theta}_t(EC_t, \varepsilon_{t+1})$ is the one which is of the same sign as $\theta_t^{U*} = \theta_{t-1} - \frac{\beta}{2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t$. For a given level of ε_{t+1} , $\bar{\theta}_t$ increases as EC_t increases. From (30) it can be seen that $\bar{\theta}_t$ is increasing in EC_t , this will cause π to be decreasing in EC_t since $\bar{\theta}_t$ is in the denominator of π . Since (30) is quadratic in $\bar{\theta}_t$, we have that $\bar{\theta}_t$ is proportional to $(EC_t)^{0.5}$, which implies that π is proportional to $(EC_t)^{-0.5}$. Given a level of θ_{t-1} , EC_t can be expressed as $EC_{t-1} + \text{constant}$, we get that π is proportional to $(EC_{t-1})^{-0.5}$. The value functions, V_U and V_L , are not functions of the available capital giving that U_C is a concave function of capital.

Notice also that $\bar{\theta}_t(EC_t, \varepsilon_{t+1} > 0) > \bar{\theta}_t(EC_t, \varepsilon_{t+1} < 0)$ when $\theta_t^U > 0$ and the weight given to the likelihood of liquidation next period is greater for a negative liquidity shock, $\pi(EC_t, \varepsilon_{t+1} > 0) < \pi(EC_t, \varepsilon_{t+1} < 0)$. Also, when $EC_t \rightarrow \infty$, $\bar{\theta}_t \rightarrow \infty$ giving $\pi \rightarrow 0$ which is consistent with the intuition that a constrained arbitrageur with a very large amount of capital behaves similar to an unconstrained arbitrageur. When $EC_t \rightarrow 0$, $\bar{\theta}_t \rightarrow 0$ giving $\pi \rightarrow 1$ which means that the zero capital arbitrageur's profits from future trading equal the cost associated with liquidating his position in the asset.

The maximization program (28), with $\pi(EC_t, \varepsilon_{t+1})$ given by (29) has a first order condition:

$$0 = -\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t - \frac{2}{\beta}(\theta_t - \theta_{t-1}) + \delta\left(\frac{\sigma_F}{\beta} - \sigma_I\right)E[\pi(EC_t, \varepsilon_{t+1})\varepsilon_{t+1}] - \frac{2\delta}{\beta}\theta_t E[\pi(EC_t, \varepsilon_{t+1})] \quad (31)$$

Solving for θ_t gives:

$$\theta_t^{C*} = \frac{\theta_{t-1} - \frac{\beta}{2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \{ \varepsilon_t - \delta E[\pi(EC_t, \varepsilon_{t+1})\varepsilon_{t+1}] \}}{1 + \delta E[\pi(EC_t, \varepsilon_{t+1})]} \quad (32)$$

To interpret the above expression let us first rewrite it in terms of the quantity traded by the arbitrageur. This gives

$$\theta_t^{C*} - \theta_{t-1} = -\frac{\beta}{2D(\varepsilon_t)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \theta_{t-1} \frac{(D(\varepsilon_t) - 1)}{D(\varepsilon_t)} + \frac{\delta\beta}{2D(\varepsilon_t)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) E[\pi(EC_t, \varepsilon_{t+1})\varepsilon_{t+1}] \quad (33)$$

where $D(\varepsilon_t) = 1 + \delta E[\pi(EC_t, \varepsilon_{t+1})] > 1$. The first term is similar to the case where the arbitrageur is unconstrained, except that we now have a $D(\varepsilon_t)$ in the denominator. Since

$D(\varepsilon_t) > 1$ this damps the quantity traded by the constrained arbitrageur. The second term contributes a component that is always of opposite sign to θ_{t-1} . The third term is also of opposite sign to θ_{t-1} because $\pi(EC_t, \varepsilon_{t+1} > 0) < \pi(EC_t, \varepsilon_{t+1} < 0)$ when $\theta_{t-1} > 0$ and $\pi(EC_t, \varepsilon_{t+1} > 0) > \pi(EC_t, \varepsilon_{t+1} < 0)$ when $\theta_{t-1} < 0$. These two terms can be interpreted as a drift towards a flat position.

Consider the case where the arbitrageur has an initial flat position, that is, $\theta_{t-1} = 0$. In this case the second term is zero. Also, $\pi(EC_t, \varepsilon_{t+1} > 0) < \pi(EC_t, \varepsilon_{t+1} < 0)$ when $\varepsilon_t < 0$ and $\pi(EC_t, \varepsilon_{t+1} > 0) > \pi(EC_t, \varepsilon_{t+1} < 0)$ when $\varepsilon_t > 0$. The intuition for this is as follows: if the current shock is negative the arbitrageur will have a long position after trading in the current period, hence a positive shock next period will increase his capital balance and lead to a lower likelihood of future liquidation. This third term also tends to bias the arbitrageur's position towards zero. Thus, when $\theta_{t-1} = 0$, the constrained arbitrageur trades a quantity that can be expressed as a linear function of the amount that the unconstrained arbitrageur trades and the quantity traded is symmetric.

If the arbitrageur's initial position is long, $\theta_{t-1} > 0$, the second and third terms are negative since $\pi(EC_t, \varepsilon_{t+1} > 0) < \pi(EC_t, \varepsilon_{t+1} < 0)$. In addition to this the denominator is greater than in the case where the arbitrageur has an initial flat position. If $\varepsilon_t < 0$, this results in the arbitrageur buying a smaller quantity of the asset than the constrained arbitrageur with a initial flat position. On the other hand, if $\varepsilon_t > 0$ this results in the arbitrageur selling a larger quantity than the constrained arbitrageur with an initial flat position. The opposite holds true when $\theta_{t-1} < 0$. The movement towards a flat position seen above has a cost associated with it by way of forgone profits, but also has the benefit of increased flexibility resulting from a lower likelihood of forced liquidation.

C. Price Patterns - Expected Deviation, Variance of Deviation and AutoCovariance of Deviation

In this sub-section we examine the impact on the price of the security by the trading of the arbitrageur. We consider three different properties of prices: (a) the expected price deviation next period, $E[I_t - P_t]$; (b) the expected squared price deviation and the variance of the price deviation, $E[(I_t - P_t)^2]$ and $Var(I_t - P_t)$; and (c) the expected serial price deviation and the auto-covariance in price deviations, $E[(I_{t+1} - P_{t+1})(I_t - P_t)]$ and $Cov(I_{t+1} - P_{t+1}, I_t - P_t)$.¹⁹ We do this for the three cases considered above: the absence of arbitrageurs, the case of the unconstrained monopolist arbitrageur and the financially constrained monopolist arbitrageur. The security price is given as $P_t = I_{t-1} + \frac{\sigma_F}{\beta} \varepsilon_t + \frac{\theta_t - \theta_{t-1}}{\beta}$, where we set the final term to zero when there are no arbitrageurs trading the security. This gives $I_t - P_t = I_t - I_{t-1} - \frac{\sigma_F}{\beta} \varepsilon_t - \frac{\theta_t - \theta_{t-1}}{\beta} = -\left(\frac{\sigma_F}{\beta} - \sigma_I\right) \varepsilon_t - \frac{\theta_t - \theta_{t-1}}{\beta}$.

1. Expected Price Deviation

In this sub-section we compare $E[I_t - P_t]$ for the three cases. In the absence of arbitrageurs $\theta_t = \theta_{t-1}$, giving $E[I_t - P_t] = -\left(\frac{\sigma_F}{\beta} - \sigma_I\right) E[\varepsilon_t] = 0$. When the arbitrageurs are

¹⁹These are the basic moments of the asset price. We could also examine the mean, variance and autocovariance of P_t , but these are similar to the expressions reported below.

unconstrained substituting for θ_t gives $E[I_t - P_t] = -\frac{1}{2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) E[\varepsilon_t] = 0$. Thus, in both these cases the expected price deviations are zero.

When a financially constrained arbitrageur trades the security, the above is no longer true. This can be seen by substituting the quantity traded from (33) into the above expression. This gives

$$\begin{aligned} E[I_t - P_t] &= -\frac{1}{2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) E \left[\frac{\varepsilon_t}{D(\varepsilon_t)} \right] + \theta_{t-1} E \left[\frac{D(\varepsilon_t) - 1}{D(\varepsilon_t)} \right] \\ &\quad - \frac{\beta\delta}{2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) E \left[\frac{\pi(E C_{t,\varepsilon_{t+1}}) \varepsilon_{t+1}}{D(\varepsilon_t)} \right] \\ &< 0 \quad \theta_{t-1} < 0 \\ &= 0 \quad \theta_{t-1} = 0 \\ &> 0 \quad \theta_{t-1} > 0 \end{aligned} \tag{34}$$

The intuition for the above result is rather simple: a financially constrained arbitrageur always trades towards a flat position to get away from the constraint. Thus, if he has a long position he sells more of the security when value-based traders want to buy it, than he buys of the security when value-based traders want to sell. For an initial long position the mean quantity that he trades is then negative. This has the effect of driving expected prices below expected value and represents a cost of the constraint. Figure 2 assumes a fixed amount of capital and plots the expected price deviation as a function of the arbitrageur's initial position.²⁰ As can be seen from the figures the bias is increasing in the initial position of the arbitrageur.

2. Expected Squared Price Deviation and Variance of Price Deviation

In this sub-section we examine two related properties of the price deviations - the expected squared deviation and the variance of the deviation.²¹ When arbitrageurs are absent from the market or when there are unconstrained arbitrageurs the expected squared deviation and the variance of the deviations are the same. This is due to the fact that the expected price deviation is zero. When a constrained arbitrageur trades the security, these two quantities are different from each other because the expected price deviation is no longer zero.

The expected squared deviation can be written as $E[(I_t - P_t)^2]$, while the variance of deviations can be written as $Var(I_t - P_t) = E[(I_t - P_t - E[I_t - P_t])^2]$. When arbitrageurs are absent from the market $E[(I_t - P_t)^2] = Var(I_t - P_t) = \left(\frac{\sigma_F}{\beta} - \sigma_I \right)^2$. Trading by unconstrained arbitrageurs reduces the variability of prices, giving $E[(I_t - P_t)^2] = Var(I_t - P_t) = \frac{1}{4} \left(\frac{\sigma_F}{\beta} - \sigma_I \right)^2$. This is 25% of the variance in the absence of the arbitrageur.

The expected squared deviation when a constrained arbitrageur trades the security is given as

²⁰The figures were plotted using both the numerical solution and the approximate analytical solution. The figures were similar. Therefore, only those generated using the numerical solution are reported in the paper.

²¹In the absence of information on the position held by the arbitrageur, the forecast of the deviation is zero. This makes the expected squared deviation, which is the unconditional variance of price deviation, the relevant measure of uncertainty.

$$E[(I_t - P_t)^2] = E \left[\left(-\left(\frac{\sigma_F}{\beta} - \sigma_I\right) \varepsilon_t + \frac{1}{2D} \left(\frac{\sigma_F}{\beta} - \sigma_I\right) \varepsilon_t + \theta_{t-1} \frac{(D-1)}{\beta D} \right)^2 \right. \\ \left. - \frac{\delta}{2D} \left(\frac{\sigma_F}{\beta} - \sigma_I\right) E[\pi(EC_t, \varepsilon_{t+1}) \varepsilon_{t+1}] \right] \quad (35)$$

As can be seen from the above equation the expected squared deviation, like the expected deviation, has an explicit dependence on the initial position of the arbitrageur.

Trading by a financially constrained arbitrageur results in a reduction in the variance of the price deviation, which is given as

$$\begin{aligned} Var(I_t - P_t) &= \left(\frac{\sigma_F}{\beta} - \sigma_I\right)^2 + 2 \left(\frac{\sigma_F}{\beta} - \sigma_I\right) E \left[\frac{\theta_t^{C*} \varepsilon_t}{\beta} \right] + \frac{E[(\theta_t^{C*})^2] - E[\theta_t^{C*}]^2}{\beta} \\ &= \left(\frac{\sigma_F}{\beta} - \sigma_I\right)^2 + \frac{1}{\beta} E[\theta_t^{C*} \varepsilon_t] \left[\left(\frac{\sigma_F}{\beta} - \sigma_I\right) + \frac{1}{4\beta} E[\theta_t^{C*} \varepsilon_t] \right] \\ &\quad \frac{1}{4} \left(\frac{\sigma_F}{\beta} - \sigma_I\right)^2 < Var(I_t - P_t) < \left(\frac{\sigma_F}{\beta} - \sigma_I\right)^2 \end{aligned} \quad (36)$$

This results from the fact that the second term is always negative since $\beta \left(\frac{\sigma_F}{\beta} - \sigma_I\right) = -E[\theta_t^{U*}(\varepsilon_t) \varepsilon_t] > -E[\theta_t^{C*}(\varepsilon_t) \varepsilon_t] > 0$.²²

Figure 3 plots the expected squared deviation as a function of the initial position of the constrained arbitrageur. From the plot it can be seen that the expected squared price deviation is increasing as the arbitrageur's initial position approaches its maximum sustainable level. Consider $\theta_{t-1} > 0$: As θ_{t-1} increases the arbitrageur sells an increasing amount of the asset to allow him to move back towards a flat position. This causes an increasing deviation in between the asset's price and value. When $\theta_{t-1} < 0$: the arbitrageur buys an increasing amount of the asset to help him move back to a flat position, causing price to deviate in the opposite direction relative to value.

3. Auto-covariance of Price Deviations

In this sub-section we again examine two related properties of the price deviations - the expected serial co-deviation $E[(I_{t+1} - P_{t+1})(I_t - P_t)]$ and the autocovariance of the price deviations, $Cov(I_{t+1} - P_{t+1}, I_t - P_t)$. In the first two cases, absence of arbitrageur and trading by an unconstrained arbitrageur, both of the above quantities are zero. In both these cases the deviations depend only on the current period shock, and the shocks are not auto-correlated giving zero expected serial co-deviation and zero autocovariance in price deviations.

The presence of a financially constrained arbitrageur induces serial correlation in the price deviations. The expected serial co-deviation and autocovariance of price deviations depend on the initial position of the arbitrageur. The expected serial co-deviation is given as

²²If $E[\theta_t^{C*}(\varepsilon_t) \varepsilon_t] = -\beta \left(\frac{\sigma_F}{\beta} - \sigma_I\right)$, the second term equals $-\frac{3}{4} \left(\frac{\sigma_F}{\beta} - \sigma_I\right)^2$, while if $E[\theta_t^{C*}(\varepsilon_t) \varepsilon_t] = 0$, the second term is 0.

$$\begin{aligned}
E_t [(I_{t+1} - P_{t+1}) (I_t - P_t)] &= E_t \left[\left(\left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_{t+1} + \frac{\theta_{t+1}^{C*} - \theta_t^{C*}}{\beta} \right) \left(\left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t + \frac{\theta_t^{C*} - \theta_{t-1}^{C*}}{\beta} \right) \right] \\
&= E_t \left[\left(\frac{\theta_{t+1}^{C*} - \theta_t^{C*}}{\beta} \right) \left(\left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t + \frac{\theta_t^{C*} - \theta_{t-1}^{C*}}{\beta} \right) \right] \\
&> 0
\end{aligned} \tag{37}$$

The intuition for this result is that the arbitrageur is always trading down towards a flat position. This will cause successive price deviations to be in the same direction. Mathematically, the autocovariance is given by

$$\begin{aligned}
Cov(I_{t+1} - P_{t+1}, I_t - P_t) &= E \left[\left(\frac{\theta_{t+1}^{C*} - E[\theta_{t+1}^{C*}]}{\beta} + \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_{t+1} \right) \left(\frac{\theta_t^{C*} - E[\theta_t^{C*}]}{\beta} + \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \right) \right] \\
&= E \left[\left(\frac{\theta_{t+1}^{C*} - E[\theta_{t+1}^{C*}]}{\beta} \right) \left(\frac{\theta_t^{C*} - E[\theta_t^{C*}]}{\beta} + \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \right) \right] \\
&> 0
\end{aligned} \tag{38}$$

Figure 4 plots the expected serial co-deviation as a function of the constrained arbitrageurs initial position in the asset. As it can be seen, the expected serial co-deviation is a quadratic function of the initial position of the constrained liquidity trader when the initial position is not very large. For traders with initial positions close to the maximum sustainable, the expected serial co-deviation rises sharply because of the high likelihood of forced liquidation. This result is consistent with the one obtained above, that the higher the likelihood of forced liquidation the greater the volatility of price deviations, thus indicating that as the arbitrageur liquidates, volatility moves up and tends to persist at high levels.

This section considered the case of a monopolist arbitrageur trading a single risky security. In the next section we examine the impact of trading by multiple financially constrained arbitrageurs competing with one another. The section following that considers the case where a single arbitrageur trades multiple risky securities.

III. Oligopolistic Competition Among Arbitrageurs

Here we consider the case where there are n symmetric arbitrageurs supplying liquidity. The market clearing condition is modified to

$$D_t^L + \theta_t - \theta_{t-1} + (n-1) \Delta_t = 0$$

where, the arbitrageur under consideration trades $\theta_t - \theta_{t-1}$ and Δ_t is the quantity traded by each of the remaining $(n-1)$ arbitrageur. The market price of the asset is given as

$$P_t = I_{t-1} + \frac{\sigma_F}{\beta} \varepsilon_t + \frac{\theta_t - \theta_{t-1}}{\beta} + \frac{(n-1) \Delta_t}{\beta}$$

In this setting, there is competition among arbitrageurs. Since it is not atomistic competition, each arbitrageur will consider the impact of his own trading as well as that of competing

arbitrageurs in determining the quantity to trade. A financially unconstrained arbitrageur optimally trades to the following position:

$$\theta_t^{UO*} = \theta_{t-1} - \frac{\beta}{(n+1)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \quad (39)$$

which solves the maximization problem after substituting for P_t into the expression for the total profits when arbitrageurs are symmetric. Assuming symmetry is not unrealistic, since asymmetry is generated by two sources: (i) the initial position of arbitrageurs and (ii) differential information or beliefs. In markets where differential information is not important, the impact of differences in the initial position will die out over time as all arbitrageurs positions will converge.

If arbitrageurs are financially constrained, we need to examine the evolution of their capital balances. If each trader accounts for the positions of the remaining traders when calculating the liquidation value of his capital, we get $RC_{t+1} = RC_t + \theta_{t-1} \left(n \frac{\theta_{t-1}}{\beta} - \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \right) - \theta_t \left(n \frac{\theta_t}{\beta} - \frac{\sigma_F}{\beta} \varepsilon_{t+1} \right)$. Let $\bar{\theta}_t$ be the limit position that the arbitrageur can hold without being forced to liquidate next period. Using $EC_t = RC_t + \theta_{t-1} \left(n \frac{\theta_{t-1}}{\beta} - \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t \right)$ we get that $\bar{\theta}_t$ solves $EC_t - \theta_t \left(n \frac{\theta_t}{\beta} - \frac{\sigma_F}{\beta} \varepsilon_{t+1} \right) = 0$. As before define weight functions $\pi(EC_t, \varepsilon_{t+1}) = \frac{a}{a + |\bar{\theta}_t(EC_t, \varepsilon_{t+1})|}$. This allows us to write the problem of a financially constrained arbitrageur as

$$U_C(EC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \begin{aligned} &Max_{\theta_t} \left\{ \begin{aligned} &-(\theta_t - \theta_{t-1}) \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{(\theta_t - \theta_{t-1})^2}{\beta} \\ &-(\theta_t - \theta_{t-1}) \frac{(n-1)\Delta_t}{\beta} \end{aligned} \right\} \\ &+ \delta E \left[\begin{aligned} &\pi(EC_t, \varepsilon_{t+1}) V_L(\theta_t, \varepsilon_{t+1}) \\ &+ (1 - \pi(EC_t, \varepsilon_{t+1})) V_U(\theta_t, \varepsilon_{t+1}) \end{aligned} \right] \end{aligned} \end{cases} \begin{aligned} &EC_{t-1} - n \frac{\theta_{t-1}^2}{\beta} > -\frac{\sigma_F}{\beta} \theta_t \\ &EC_{t-1} - n \frac{\theta_{t-1}^2}{\beta} \leq -\frac{\sigma_F}{\beta} \theta_t \end{aligned} \end{cases} \quad (40)$$

where, $V_L(\theta_t, \varepsilon_{t+1}) = \theta_t \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_{t+1} - \frac{n\theta_t^2}{\beta}$, which is obtained by setting $\Delta_{t+1} = -\theta_t$ because of simultaneous forced liquidation at $t+1$, and $V_U(\theta_t, \varepsilon_{t+1}) = \frac{\beta}{(n+1)^2} \left(\frac{\sigma_F}{\beta} - \sigma_I \right)^2 \left(\varepsilon_{t+1}^2 + \frac{\delta}{1-\delta} \right)$. Solving for the optimal position gives

$$\theta_t^{CO*} = \frac{\theta_{t-1} - \frac{\beta}{(n+1)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \{ \varepsilon_t - \delta E [\pi(EC_t, \varepsilon_{t+1}) \varepsilon_{t+1}] \}}{\left(1 + \frac{2\delta n}{(n+1)} E [\pi(EC_t, \varepsilon_{t+1})] \right)} \quad (41)$$

It is not surprising that the financially constrained oligopolistic arbitrageur holds a smaller position in the asset than if he were unconstrained. Also, the difference between the oligopolist case and the monopolist case is the standard one. Individually, each competitive arbitrageur holds less than the monopolist arbitrageur, but on aggregate the quantity traded by oligopolistic arbitrageurs is larger than that of the monopolist, and therefore the impact on market prices is higher under the oligopoly, and the profit per unit and the total arbitrage profits are also lower.

From the above it can be seen that the oligopolist arbitrageur's optimal quantity traded is determined in a similar way to the monopolist arbitrageur's. The results on price patterns are also very similar in this case.

The above results were obtained assuming that the arbitrageurs held the same initial positions and had the same amount of capital available to them. A situation that may sometimes be observed is one where the arbitrageurs have different amounts of capital available to them or/and have different initial positions. The discussion here is an attempt to understand how the above results would change in this setting.

We can consider the heterogeneity of capital availability and initial positions together since the quantity that is important is the initial position relative to the maximum sustainable. Let n arbitrageurs trade an asset. As seen above, the quantity traded by constrained arbitrageurs is made up of two parts - the purpose of one component of the trade is to help move the arbitrageur's position down to zero, the second component is the supply of liquidity that generates trading profits.

First consider the case where the total initial position held by the competing arbitrageurs is close to zero. In this setting the arbitrageurs start out with initial positions on both sides of the market - some have long positions in the asset while others have short positions. Each arbitrageur will be aggressive in determining the magnitude of the first component of the trade since the cost of moving towards a flat position will be relatively low - competing arbitrageurs will be on different sides. This means that situations with arbitrageurs on both sides of the market with total initial position close to zero will tend to disappear rather quickly. The second component of the trade, the supply of liquidity will also be determined more aggressively than in the symmetric case, though much less aggressively than the first component of the trade. The aggressiveness in the determination of the second component of the trade is lower since the arbitrageurs would not rationally expect the initial conditions to persist.

Now, let us consider a second case where the arbitrageurs have positions of various sizes but the total position of the arbitrageurs is significantly different from zero, say a net long position. In this setting the total of the first component of the trade across arbitrageurs will be negative. Arbitrageurs who have an initial short position will be able to aggressively liquidate their positions, while arbitrageurs with initial long positions will be forced to be much less aggressive in their movement towards a flat position. On the second component of the trade, the arbitrageurs with a greater level of flexibility, those with smaller initial positions relative to the amount of capital available will be able to supply liquidity more aggressively allowing them to earn greater profits.

We next consider the case of an arbitrageur who can trade multiple assets.

IV. Arbitrage in Multiple Assets

In this section we consider the problem of a monopolist arbitrageur who can trade multiple risky assets. For simplicity, we only consider two assets. We assume that the links between the markets occur through the trading of the arbitrageur, the shocks to the values of the assets and to the liquidity demand in the various markets. If the arbitrageur is unconstrained

his optimization problem is similar to that of two arbitrageurs each trading a single asset. This allows us to write the value function for the two-asset unconstrained problem as

$$V_U(\theta_{1,t}, \theta_{2,t}, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) = \sum_{i=1}^2 \frac{\beta_i}{4} \left(\frac{\sigma_{iF}}{\beta_i} - \sigma_{iI} \right)^2 \left[\varepsilon_{i,t+1}^2 + \frac{\delta}{1-\delta} \right] \quad (42)$$

Consider next how the financially constrained monopolist arbitrageur splits his trades when there two assets. Position limits are computed from the capital available which depends on the portfolio holdings:

$$RC_{t+1} = RC_t + \sum_{i=1}^2 \frac{1}{\beta_i} \theta_{i,t-1} \left(\theta_{i,t-1} - \beta_i \left(\frac{\sigma_{iF}}{\beta_i} - \sigma_{iI} \right) \varepsilon_{i,t} \right) - \sum_{i=1}^2 \frac{1}{\beta_i} \theta_{i,t} \left(\theta_{i,t} - \sigma_{i,F} \varepsilon_{i,t+1} \right) \quad (43)$$

Define the weight function $\pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) = \frac{a}{a + |\bar{\theta}_{1t}(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1})|}$ where $\bar{\theta}_{1t}(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1})$ is obtained by setting to zero the amount of capital left after liquidating the portfolio next period, $EC_t - \sum_{i=1}^2 \frac{1}{\beta_i} \theta_{i,t} (\theta_{i,t} - \sigma_{i,F} \varepsilon_{i,t+1}) = 0$. As before this is obtained by using $EC_t = RC_t + \sum_{i=1}^2 \frac{1}{\beta_i} \theta_{i,t-1} \left(\theta_{i,t-1} - \beta_i \left(\frac{\sigma_{iF}}{\beta_i} - \sigma_{iI} \right) \varepsilon_{i,t} \right)$ in the above expression for RC_{t+1} . We assume that if the arbitrageur is not forced into liquidation he will trade to maintain next period a ratio of holdings, $\frac{\theta_{2,t}}{\theta_{1,t}} = \frac{\theta_{2,t}^U}{\theta_{1,t}^U} = \alpha_t$ ²³. $\bar{\theta}_{1t}(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1})$ then solves

$$0 = EC_t - \left(\frac{1}{\beta_1} + \frac{\alpha_t^2}{\beta_2} \right) \bar{\theta}_{1,t}^2 + \left(\frac{\sigma_{1,F}}{\beta_1} \varepsilon_{1,t+1} + \alpha_t \frac{\sigma_{2,F}}{\beta_2} \varepsilon_{2,t+1} \right) \bar{\theta}_{1,t}. \quad (44)$$

The objective function can then be written as

$$\begin{aligned} & \underset{\theta_{1,t}, \theta_{2,t}}{Max} \sum_{i=1}^2 \left[-(\theta_{i,t} - \theta_{i,t-1}) \left(\frac{\sigma_{iF}}{\beta_i} - \sigma_{iI} \right) \varepsilon_{i,t} - \frac{(\theta_{i,t} - \theta_{i,t-1})^2}{\beta_i} \right] \\ & + \delta E \left[\begin{aligned} & \pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) V_L(\theta_{1,t}, \theta_{2,t}, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) \\ & + (1 - \pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1})) V_U(\theta_{1,t}, \theta_{2,t}, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) \end{aligned} \right] \end{aligned} \quad (45)$$

and the value of the trading strategy when the arbitrageur is forced to liquidate because of zero capital is also similar to the single asset example:

$$V_L(\theta_{1,t}, \theta_{2,t}, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) = \sum_{i=1}^2 \left[\theta_{i,t} \left(\frac{\sigma_{iF}}{\beta_i} - \sigma_{iI} \right) \varepsilon_{i,t+1} - \frac{\theta_{i,t}^2}{\beta_i} \right]$$

The solution to the constrained optimization problem for asset 1 is

$$\theta_{1,t}^{C*} = \frac{\theta_{1,t-1} - \frac{\beta_1}{2} \left(\frac{\sigma_{1,F}}{\beta_1} - \sigma_{1,I} \right) (\varepsilon_{1,t} - \delta E[\pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) \varepsilon_{1,t+1}])}{(1 + \delta E[\pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1})])}$$

²³The expression for the evolution of the capital has two variables, $\theta_{1,t}$ and $\theta_{2,t}$. This assumption allows us to fix the value of one variable relative to the other. In the single asset case θ_t is the maximum position that can be sustained without triggering bankruptcy next period. In the two asset case we would have to solve for $\bar{\theta}_{1,t}$ and $\bar{\theta}_{2,t}$, the maximum in each of the two assets, where $\bar{\theta}_{1,t}$ and $\bar{\theta}_{2,t}$ would be related. As a result of the assumption we consider only a particular combination of $\bar{\theta}_{1,t}$ and $\bar{\theta}_{2,t}$. The combination that we consider has the same ratio of holdings as held by the unconstrained arbitrageur.

and similarly for asset 2,

$$\theta_{2,t}^{C*} = \frac{\theta_{2,t-1} - \frac{\beta_2}{2} \left(\frac{\sigma_{2,F}}{\beta_2} - \sigma_{2,I} \right) (\varepsilon_{2,t} - \delta E[\pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1}) \varepsilon_{2,t+1}])}{(1 + \delta E[\pi(EC_t, \varepsilon_{1,t+1}, \varepsilon_{2,t+1})])}$$

Notice that the above solutions do not contain an explicit dependence of the quantities traded of one asset on the direction of the shock to the other asset. The dependence results from the expressions for the π 's, which contain the shocks and the holdings of both assets. The arbitrageur's capital balance changes with the direction of the shock and the current holdings of both assets, which affects $\bar{\theta}_t$ and this in turn affects the weights used to approximate future profits. Thus, the quantity of each asset traded depends on the liquidity shock to the other asset as well as the position held on the other asset. Thus we find that financially constrained arbitrageurs do not sell what they want but what they can.

A. Price Patterns - Expected Cross Deviation

In this sub-section we examine the expected cross deviation, $E[(I_{1t} - P_{1t})(I_{2t} - P_{2t})]$, in the two asset prices. In the absence of the arbitrageur the two asset prices are linked only by the fact that the liquidity and value shocks are correlated. Using the expressions for the price of the assets from section 2 we can write

$$\begin{aligned} E[(I_{1t} - P_{1t})(I_{2t} - P_{2t})] &= \left(\frac{\sigma_{1F}}{\beta_1} - \sigma_{1I} \right) \left(\frac{\sigma_{2F}}{\beta_2} - \sigma_{2I} \right) E[\varepsilon_{1,t} \varepsilon_{2,t}] \\ &= \left(\frac{\sigma_{1F}}{\beta_1} - \sigma_{1I} \right) \left(\frac{\sigma_{2F}}{\beta_2} - \sigma_{2I} \right) \rho \end{aligned} \quad (46)$$

where, $E[\varepsilon_{1,t} \varepsilon_{2,t}] = \rho$ is the correlation of liquidity shocks across markets. The presence of an unconstrained arbitrageur does not change the results significantly. There is a reduction the magnitude, but this results purely from the fact that the arbitrageur absorbs part of the liquidity shock. We can now write

$$E[(I_{1t} - P_{1t})(I_{2t} - P_{2t})] = \frac{1}{4} \left(\frac{\sigma_{1F}}{\beta_1} - \sigma_{1I} \right) \left(\frac{\sigma_{2F}}{\beta_2} - \sigma_{2I} \right) \rho \quad (47)$$

This reduction in the magnitude of the expected cross deviation is similar to the reduction in the magnitude of the variance of the deviation in the single asset unconstrained case. If we standardize the above expressions by the standard deviation of the price deviation, we obtain the correlation of the price deviation, which in both cases is equal to ρ . Thus, the absence of an arbitrageur or trading by an unconstrained arbitrageur does not affect the correlation between the asset prices.

This is no longer true if there is a constrained arbitrageur trading the assets. In this case, the expected cross deviation depends on the arbitrageurs holdings of the two assets. The expression for the expected cross deviation is given as

$$\begin{aligned}
E[(I_{1t} - P_{1t})(I_{2t} - P_{2t})] &= E\left[\left(\left(\frac{\sigma_{1F}}{\beta_1} - \sigma_{1I}\right)\varepsilon_{1t} + \frac{(\theta_{1t}^{C*} - \theta_{1t-1})}{\beta_1}\right)\left(\left(\frac{\sigma_{2F}}{\beta_2} - \sigma_{2I}\right)\varepsilon_{2t} + \frac{(\theta_{2t}^{C*} - \theta_{2t-1})}{\beta_2}\right)\right] \\
&= \left(\frac{\sigma_{1F}}{\beta_1} - \sigma_{1I}\right)\left(\frac{\sigma_{2F}}{\beta_2} - \sigma_{2I}\right)\rho + E\left[\left(\frac{\sigma_{1F}}{\beta_1} - \sigma_{1I}\right)\frac{(\theta_{2t}^{C*} - \theta_{2t-1})}{\beta_2}\varepsilon_{1t}\right] \\
&\quad + E\left[\left(\frac{\sigma_{2F}}{\beta_2} - \sigma_{2I}\right)\frac{(\theta_{1t}^{C*} - \theta_{1t-1})}{\beta_1}\varepsilon_{2t}\right] + E\left[\frac{(\theta_{2t}^{C*} - \theta_{2t-1})}{\beta_2}\frac{(\theta_{1t}^{C*} - \theta_{1t-1})}{\beta_1}\right]
\end{aligned} \tag{48}$$

To interpret this equation, let us re-write the expressions of the quantity that the arbitrageur trades as

$$\begin{aligned}
\theta_{1,t}^{C*} - \theta_{1,t-1} &= -\frac{\beta_1}{2D}\left(\frac{\sigma_{1,F}}{\beta_1} - \sigma_{1,I}\right)\varepsilon_{1,t} - \theta_{1,t-1}\frac{(D-1)}{D} \\
&\quad + \frac{\delta\beta_1}{2D}\left(\frac{\sigma_{1,F}}{\beta_1} - \sigma_{1,I}\right)E[\pi(\varepsilon_{1,t+1}, \varepsilon_{2,t+1})\varepsilon_{1,t+1}]
\end{aligned}$$

and

$$\begin{aligned}
\theta_{2,t}^{C*} - \theta_{2,t-1} &= -\frac{\beta_2}{2D}\left(\frac{\sigma_{2,F}}{\beta_2} - \sigma_{2,I}\right)\varepsilon_{2,t} - \theta_{2,t-1}\frac{(D-1)}{D} \\
&\quad + \frac{\delta\beta_2}{2D}\left(\frac{\sigma_{2,F}}{\beta_2} - \sigma_{2,I}\right)E[\pi(\varepsilon_{1,t+1}, \varepsilon_{2,t+1})\varepsilon_{2,t+1}]
\end{aligned}$$

where, $D = [1 + \delta E[\pi(\varepsilon_{1,t+1}, \varepsilon_{2,t+1})]] > 1$.

For the remainder of the discussion in this section assume that $\sigma_{1F} = \sigma_{2F} = \sigma_F$, $\sigma_{1I} = \sigma_{2I} = \sigma_I$, $\rho = 0$ and $\beta_1 = \beta_2 = \beta$. Consider first the case when $\theta_{1,t-1} = \theta_{2,t-1} = 0$. In this case, the arbitrageur trades the two assets symmetrically and the expected cross deviation is zero²⁴. The expected cross deviation is also zero if either $\theta_{1,t-1} = 0$ or $\theta_{2,t-1} = 0$. In these cases there is zero price deviation for one of the assets, while there is a price deviation for the other asset, giving an expected cross deviation of zero.

Next, consider the case where $\theta_{1,t-1}$ and $\theta_{2,t-1}$ are of the same sign, that is, the arbitrageur holds initial long positions in both assets or he holds initial short positions in both assets. Assume that he has long positions in both the assets. In this case, part of the arbitrageurs trade is directed towards trading down to a flat position in both assets. For any magnitude of the liquidity shock to the two assets the arbitrageur always sells more of the asset if the shock is positive than he buys of the asset if the shock is negative. This results in an expected cross deviation across the two assets that is positive. The same results when the arbitrageur holds initial short positions in the two assets. The expected cross deviation is highest when the positions in the two assets are equal.

A similar line of reasoning gives the result that the expected cross deviation is negative when the arbitrageur is initially long one asset and short the other. In this case the arbitrageur in an attempt to trade down to a flat position is biasing his trade in the asset that he is long towards selling and towards buying in the asset that he is short. This gives an expected cross deviation in the assets that is negative.

Figure 5 plots the expected cross deviation of price as a function of the initial position in the two assets. The horizontal axis is a measure of the relative position in the two assets,

²⁴Let $z_1, z_2 \in (0, \infty)$. The set of liquidity shocks to the two assets can be (z_1, z_2) , $(-z_1, z_2)$, $(z_1, -z_2)$ and $(-z_1, -z_2)$. If the correlation between the shocks is zero, the four outcomes are equally probable. For $\theta_{1,t-1} = \theta_{2,t-1} = 0$, we have $\theta_{1,t}(-z_1, z_2) = \theta_{1,t}(-z_1, -z_2) = -\theta_{1,t}(z_1, z_2) = -\theta_{1,t}(z_1, -z_2)$ and $\theta_{2,t}(z_1, z_2) = \theta_{2,t}(-z_1, z_2) = -\theta_{2,t}(-z_1, -z_2) = -\theta_{2,t}(z_1, -z_2)$. This implies that the expected cross-deviations computed over these four points is zero. Integrating over all possible combinations of (z_1, z_2) gives that the expected cross deviation is zero.

$\tan^{-1} \left(\frac{\theta_{2,t-1}}{\theta_{1,t-1}} \right)$ and goes from 0 to 2π . The case where $\theta_{1,t-1} > 0$ and $\theta_{2,t-1} > 0$ is from 0 to $\frac{\pi}{2}$, $\theta_{1,t-1} < 0$ and $\theta_{2,t-1} > 0$ is from $\frac{\pi}{2}$ to π , $\theta_{1,t-1} < 0$ and $\theta_{2,t-1} < 0$ is from π to $\frac{3\pi}{2}$ and $\theta_{1,t-1} > 0$ and $\theta_{2,t-1} < 0$ is from $\frac{3\pi}{2}$ to 2π . The three lines in the graph correspond to initial positions, $\sqrt{\theta_{1,t-1}^2 + \theta_{2,t-1}^2}$, that are 30%, 60% and 90% of maximum sustainable.

As can be seen from Figure 5 the expected cross deviation is positive when the initial positions are of the same sign, between 0 and $\frac{\pi}{2}$ and π and $\frac{3\pi}{2}$, and negative when the initial positions are of opposite signs, between $\frac{\pi}{2}$ and π and $\frac{3\pi}{2}$ and 2π . The line with the smallest variation corresponds to the case where the total initial position is 30% of maximum sustainable, while the line with the greatest variation corresponds to the case where the initial position is 90% of maximum sustainable. Correlations between asset returns are therefore dependent on the arbitrageurs' financial health and holdings. Events in one market that have a negative impact in the arbitrageurs' capital balances lead to the re-arrangement of positions in all other assets. Markets then become more closely linked and the correlations among seemingly unrelated assets move towards unity.

V. Implications for the Regulation of Arbitrage Activities

In the last decade banks have greatly increased their holdings of traded assets. The increase in the relative importance of trading risk in bank portfolios poses new challenges to regulators. In 1996 the Basle Committee addressed the issue by requiring banks to hold capital equivalent to a percentage of their holdings in different asset categories, where the percentages are set to reflect the price volatilities of standard assets in the relevant categories. The benchmark used to define risk levels, value at risk (*VaR*) is a cut-off level that the loss over a given time horizon exceeds with some probability. The Bank for International Settlements (BIS), for example, sets this probability to 1% over a ten day period for purposes of determining the adequacy of bank capital. Banks themselves have developed value at risk measures for internal purposes, such as the J.P. Morgan's daily *VaR* at the 5% probability (95% confidence level).

Whether the *VaR* of a firm's portfolio of positions is a good measure of the risk of financial distress over a short period depends on the liquidity of the portfolio, as well as on the risk of extreme cash outflows or on severe disruptions to market liquidity. Market liquidity is indeed a relevant precondition for *VaR* measures of risk to work, since in markets that lack the necessary depth, traders may face difficulties in rearranging their positions. In more extreme circumstances, attempts to trade in a temporarily shallow market to ensure compliance with regulations may impose large losses on traders' capital, which is exactly what regulations are supposed to help prevent. Capital adequacy requirements which appear to work under normal circumstances then fail at times when they are most needed.

VaR measures currently in use seem to ignore that trading by large financial institutions can have an decisive impact on market prices in less than perfectly liquid markets. The common assumption in existing *VaR* models that price volatilities are exogenous is problematic for two reasons. First, *VaR* incorrectly measures the extent of potential forced reductions of the bank's capital at some confidence level. Second, it creates distortions to social welfare and competition. We show that current *VaR* models underestimate the required capital

for large banks at the expense of overestimating the amount of capital required for other banks. The problem lies in the existing regulatory benchmarks, which ignore the fact that the actions of regulated firms affect the distribution of assets returns, which serve as the basis for setting the minimum capital requirements. This is especially critical in the case of large banks, for which sound oversight is all the more important to safeguard the financial system.

VaR models rely on the specification of random changes in the market prices of the assets and on a model for computing the sensitivity of the bank's capital to prices of the underlying securities held. To highlight the importance of building *VaR* models that correctly reflect trading risk of financially constrained institutions, we will separate the impact of financial constraints from the impact of the assumption that arbitrageurs do not effect prices when they execute their transactions.

Consider the case of the unconstrained arbitrageur presented in Section II.A and assume that the innovations are normally distributed with mean zero and unit variance, that is $\varepsilon_t \sim N(0, 1)$. Gains and losses from trading are given by $\theta_{t-1}(P_t - P_{t-1})$, where P_t is normally distributed with mean I_{t-1} and variance $\frac{1}{4}(\frac{\sigma_F}{\beta} + \sigma_I)^2$. These gains and losses are also normally distributed, $N\left(\theta_{t-1}(I_{t-1} - P_{t-1}), \frac{\theta_{t-1}^2}{4}\left(\frac{\sigma_F}{\beta} + \sigma_I\right)^2\right)$. The *VaR* is then simply given as

$$VaR = -\theta_{t-1} \left((I_{t-1} - P_{t-1}) + \frac{z_n}{2} \left(\frac{\sigma_F}{\beta} + \sigma_I \right) \right) \quad (49)$$

where $z_n < 0$ is the number of standard deviations from the mean that gives a loss in capital that is exceeded with probability n over a period of time.

In the case of the constrained arbitrageur in Section II.B, gains and losses are computed using θ_t^C from (32) in the equation for P_t . Substituting gives an expression for the gains/losses:

$$\theta_{t-1} \left(+ \frac{\delta}{2D(\varepsilon_t)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) E \left[\pi(EC_t, \varepsilon_{t+1}) \varepsilon_{t+1} \right] + \left[\frac{\sigma_F}{\beta} - \frac{1}{2D(\varepsilon_t)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \right] \varepsilon_t \right) \quad (50)$$

and the corresponding *VaR* measure that obtains is:

$$VaR = -\theta_{t-1} \left(+ \frac{\delta}{2D(z_n)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) E \left[\pi(EC_t, \varepsilon_{t+1}) \varepsilon_{t+1} \right] + \left[\frac{\sigma_F}{\beta} - \frac{1}{2D(z_n)} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \right] z_n \right) \quad (51)$$

Consider now that the regulators in setting the minimum capital requirements overlook the fact that arbitrageurs' trades impact asset prices. These prices are then assumed to be given by $P_t = I_t + \frac{\sigma_F}{\beta} \varepsilon_t$ and in turn the gains and losses are assumed to be: $\theta_{t-1}(P_t - P_{t-1}) = \theta_{t-1} \left((I_{t-1} - P_{t-1}) + \frac{\sigma_F}{\beta} \varepsilon_t \right)$. The amount of capital required that satisfies the *VaR* at the n per cent level of confidence is then set equal to

$$MC_{t-1} \geq -M\theta_{t-1} \left((I_{t-1} - P_{t-1}) + z_n \frac{\sigma_F}{\beta} \right)$$

where M is an arbitrary coverage ratio and MC_t is the capital computed at market prices under the assumption that trading by arbitrageurs does not affect prices. Note that the amount of capital is effectively like a margin requirement with a variable component and a fixed component. The first component depends on the price deviations assumed under no influence from arbitrageurs, and the fixed component depends both on a level of price volatility, assuming that arbitrage trading does not contribute to this volatility, and on the value-based traders' demands. The only reason why this component is fixed lies on the incorrect assumption that arbitrageurs' trading strategies do not impact prices, and therefore it can be exogenously determined.²⁵

Financially constrained arbitrageurs take this VaR measure in their capital requirements when deciding on their optimal trading strategies, knowing that their trades impact on prices. The optimization problem in Section II.B remains the same, but where the requirement of positive regulatory capital is replaced by an equivalent capital adequacy constraint, $MC_{t-1} \geq -M\theta_{t-1}[(I_{t-1} - P_{t-1}) + z_n \frac{\sigma_F}{\beta}]$. Expression (16) is then replaced by

$$V_C(MC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \underset{\theta_t}{Max} \quad \theta_{t-1}\sigma_I\varepsilon_t - \frac{(\theta_t - \theta_{t-1})^2}{\beta} & MC_t > VaR_t \\ \quad -(\theta_t - \theta_{t-1})\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t \\ \quad + \delta E[V_C(MC_t, \theta_t, \varepsilon_{t+1})] & \\ \theta_{t-1}\sigma_I\varepsilon_t + \theta_{t-1}\left(\frac{\sigma_F}{\beta} - \sigma_I\right)\varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & MC_t \leq VaR_t \end{cases} \quad (52)$$

Solving for the maximum position that the trader can hold in the risky asset without violating the capital adequacy constraint, gives now $\bar{\theta}_t(MC_t, \varepsilon_{t+1}) = MC_t/M(I_t - P_t + \frac{\sigma_F}{\beta}\varepsilon_{t+1})$, which can be seen to differ from (30). The optimal holding in the asset remains as the solution to (28), but where the weight π is now different.

When regulators ignore the impact of arbitrageurs actions on the level and the volatility of prices the maximum sustainable position is a linear function of the capital required to stay in business. If adopted, the correct capital requirement should produce results consistent with those shown in Figure 6. The VaR based position limits/capital requirement are represented by the straight lines in the figure. The concavity of the correct function means that it lies above the linear function implied by the standard VaR capital adequacy rule that ignores the effects of arbitrage trading for small levels of capital, and it lies below the linear function for large levels of capital. What this means is that standard VaR rules force small institutions, those with lower levels of capital, to take less aggressive positions than they could if the correct measure was used. Similarly, standard VaR rules allow large institutions, those with higher levels of capital, to take more aggressive positions than they could if the correct capital requirement measure was used. This creates inefficiencies in the financial system. The social costs created by the unexpected default of a large institution, while it

²⁵It is possible to see that the capital required to stay in business depends only on the previous two shocks, after replacing both P_{t-1} and I_{t-1} by their corresponding values:

$$MC_{t-1} \geq -M\theta_{t-1} \left(\left(\sigma_I\varepsilon_{t-2} - \frac{\sigma_F}{\beta}\varepsilon_{t-1} \right) + z_n \frac{\sigma_F}{\beta} \right)$$

still seems solvent under the standard VaR measure, are much greater than those resulting from the default of a small arbitrageur. The effects on competition are also important, since small arbitrageurs can justifiably complain that they are not in a level playing field.

The analysis performed here helps to see why a system of capital requirements that ignores that arbitrageurs affect prices has clear drawbacks in its treatment of trading risk. If after a negative shock, financially fragile securities firms are forced to liquidate positions to comply with regulatory requirements, markets can become more volatile and this can contribute to further losses in the value of the arbitrageurs' portfolios. This is also the case when a cash margin call precipitates sales which further depress prices. The intention of creating a stable environment that might have motivated the idea of capital adequacy standards is then put into question. This is of even greater importance since events in one market that affect the wealth of financially constrained arbitrageurs may be transmitted to other markets where nothing has fundamentally changed. In these situations what may have been considered a low risk arbitrage opportunity, due to the low correlations of the asset returns, can turn into a very risky investment. The correlations instead of being exogenously determined as assumed are dependent on the state of the arbitrageurs' financial health and events in one market that negatively impact the arbitrageurs capital may drive the correlations towards unity.

VI. Conclusion

We have developed a dynamic arbitrage strategy which takes into account the impact of arbitrage on prices, thus avoiding the need of risk aversion or position limits to prevent infinite arbitrage positions. The stochastic process for the asset price deviations has been endogenized in a way that appears to be consistent with empirical evidence.

The model consists of stochastic demand by value-based traders that induces noise and trading by arbitrageurs that modifies the mispricing. The results demonstrate the importance of different competitive structures in the arbitrage market. They also suggest that financially constrained arbitrageurs may contribute to excess volatility in prices, and that they are responsible for the variation in the correlation of asset prices across different markets. In some cases arbitrageurs themselves contribute to the propagation of turbulence.

The findings also raise issues that are important from a regulatory view point. Specifically, it shows that measures that are used to determine the risk of arbitrage capital are not exogenous and are themselves influenced by the actions of arbitrageurs

Appendix: Optimal Consumption and Trading Policies

We have, from (9) that, the Economic Capital is given as,

$$EC_t = \theta_{t-1} I_t - B_t, \quad (53)$$

while, from (11) that, the regulatory capital is given as

$$RC_t = \theta_{t-1} P_t (\theta_t = 0, \varepsilon_t) - B_t \quad (54)$$

allowing us to write $EC_t = RC_t + \theta_{t-1} (I_t - P_t (\theta_t = 0, \varepsilon_t))$. Let C_t denote the consumption in period t . As before assume that C_t is the amount in excess of a capital charge of rEC_{t-1} . The change in economic capital from time $t-1$ to time t is given by

$$EC_t = EC_{t-1} + ((\theta_{t-1} - \theta_{t-2}) (I_{t-1} - P_{t-1}) - C_{t-1}) (1 + r) + \theta_{t-1} (I_t - I_{t-1}) \quad (55)$$

where, $r < \frac{1}{\delta} - 1$, is the rate of return on assets. Using the expression linking regulatory capital to economic capital and the evolution of economic capital we get

$$RC_t = EC_{t-1} + ((\theta_{t-1} - \theta_{t-2}) (I_{t-1} - P_{t-1}) - C_{t-1}) (1 + r) - \theta_{t-1} (I_{t-1} - P_t (\theta_t = 0, \varepsilon_t)). \quad (56)$$

The condition that $RC_t > 0$ can be written for $\theta_{t-1} > 0$ ($\theta_{t-1} < 0$) as $\varepsilon_t > L_t$ ($\varepsilon_t < L_t$) where

$$L_t = -\frac{X_{t-1}}{\theta_{t-1}} \frac{\beta}{\sigma_F} + \frac{\theta_{t-1}}{\sigma_F} \quad (57)$$

with $X_{t-1} = EC_{t-1} + ((\theta_{t-1} - \theta_{t-2}) (I_{t-1} - P_{t-1}) - C_{t-1}) (1 + r)$ and

$$X_t = X_{t-1} + \theta_{t-1} (I_t - I_{t-1}) + ((\theta_t - \theta_{t-1}) (I_t - P_t) - C_t) (1 + r). \quad (58)$$

The arbitrageur's problem can be written as

$$V_C (RC_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \max_{C_t, \theta_t} C_t + \delta E [V_C (RC_t, \theta_t, \varepsilon_{t+1})] & RC_t > 0 \\ V_L (\theta_{t-1}, \varepsilon_t) & RC_t \leq 0 \end{cases} \quad (59)$$

where, $V_L (\theta_{t-1}, \varepsilon_t) = \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta}$, the cost of liquidating the arbitrageur's position is obtained by setting θ_t to zero. Which for $\theta_{t-1} > 0$ simplifies to (for $\theta_{t-1} < 0$ the liquidation and continuation regions are switched around)

$$U_C (X_{t-1}, \theta_{t-1}, \varepsilon_t) = \begin{cases} \max_{C_t, \theta_t} C_t + \delta E [U_C (X_t, \theta_t, \varepsilon_{t+1})] & \varepsilon_t > -\frac{X_{t-1}}{\theta_{t-1}} \frac{\beta}{\sigma_F} + \frac{\theta_{t-1}}{\sigma_F} \\ \theta_{t-1} \left(\frac{\sigma_F}{\beta} - \sigma_I \right) \varepsilon_t - \frac{\theta_{t-1}^2}{\beta} & \varepsilon_t \leq -\frac{X_{t-1}}{\theta_{t-1}} \frac{\beta}{\sigma_F} + \frac{\theta_{t-1}}{\sigma_F} \end{cases} \quad (60)$$

where $U_C (X_{t-1}, \theta_{t-1}, \varepsilon_t)$ is equivalent to $V_C (RC_{t-1}, \theta_{t-1}, \varepsilon_t)$.

Consider the current period. The cut-off between the continuation region and the liquidation region is similar to that in the problem considered in Section IIb. Suppose that the liquidity shock is such that the arbitrageur can continue trading. If the arbitrageur has a small position relative to capital, the arbitrageur will have an incentive to consume an amount greater than trading profit. This is true because the marginal increase in the

bankruptcy probability will be small in this case and the arbitrageur bears a cost for holding too much capital since he discounts future utility at a rate higher than the return on investment. Thus, this situation is unlikely to persist and we are likely to observe more aggressive arbitrage, capital and positions better matched, as a result. Since the arbitrageurs are likely to operate closer to the constraint, the constraint will impact on their trades more frequently. The impact of the constraint on the quantity traded will be smaller than before since the arbitrageur has a second control variable, consumption, that he can use to reduce the risk of future bankruptcy. Because of this the likelihood of bankruptcy is not eliminated, and may not even be significantly reduced, as a result of allowing the arbitrageur to optimally pick consumption.

Figures

The liquidity shock, ε_t , is observed	A solvency check is made; Is $RC_t > 0$? If solvent the arbitrageur in control. If insolvent, lenders take control.	If solvent, trade $(\theta_t - \theta_{t-1})$, price is P_t , intrinsic value (cum-dividend) is I_t . If insolvent liquidate	Dividends are received, $\theta_t d_t$, intrinsic value is known.	Arbitrageur consumes his trading profit, $(\theta_t - \theta_{t-1})(I_t - P_t)$
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Figure 1: The sequence of events each period.

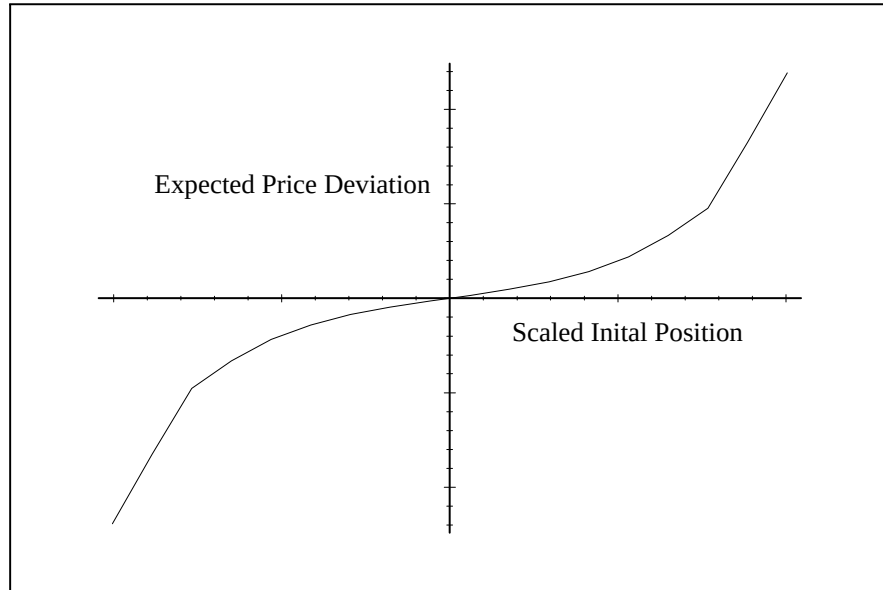


Figure 2: **Expected Price Deviation**, $E[I_t - P_t]$, as a function of the initial position. The horizontal axis is the initial position scaled by the maximum position that can be held by an arbitrageur who is willing to bear a 1% probability of bankruptcy in a given period. The following parameter values were used: $\sigma_F = 0.15$, $\sigma_I = 0.1$, $\beta = 1.0$ and the figure was generated using the exact numerical solution. A figure, not reported, was also generated using the approximation and was similar in shape and magnitude.

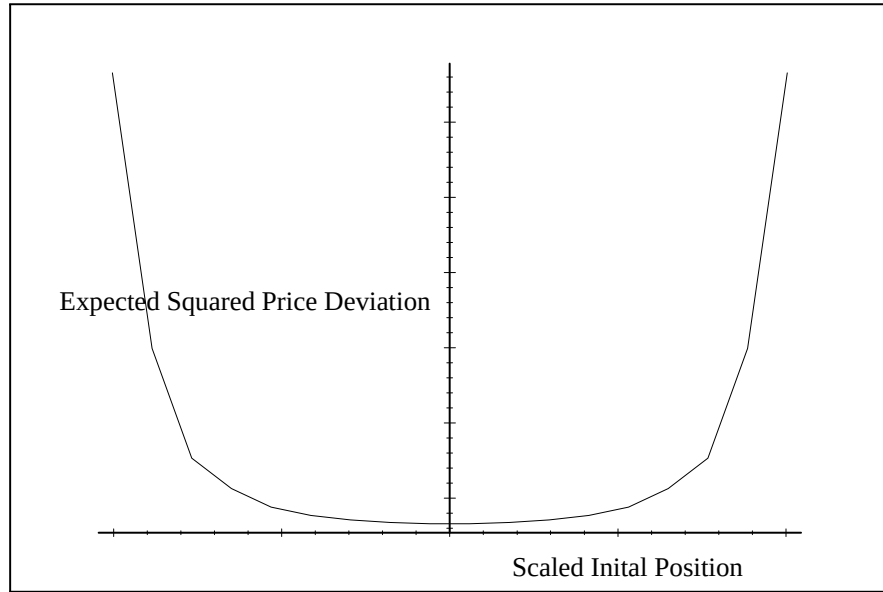


Figure 3: **Expected Squared Price Deviation** as a function of the initial position. The horizontal axis is the initial position scaled by the maximum position that can be held by an arbitrageur who is willing to bear a 1% probability of bankruptcy in a given period. The following parameter values were used: $\sigma_F = 0.15$, $\sigma_I = 0.1$, $\beta = 1.0$ and the figure was generated using the exact numerical solution. A figure, not reported, was also generated using the approximation and was similar in shape and magnitude.

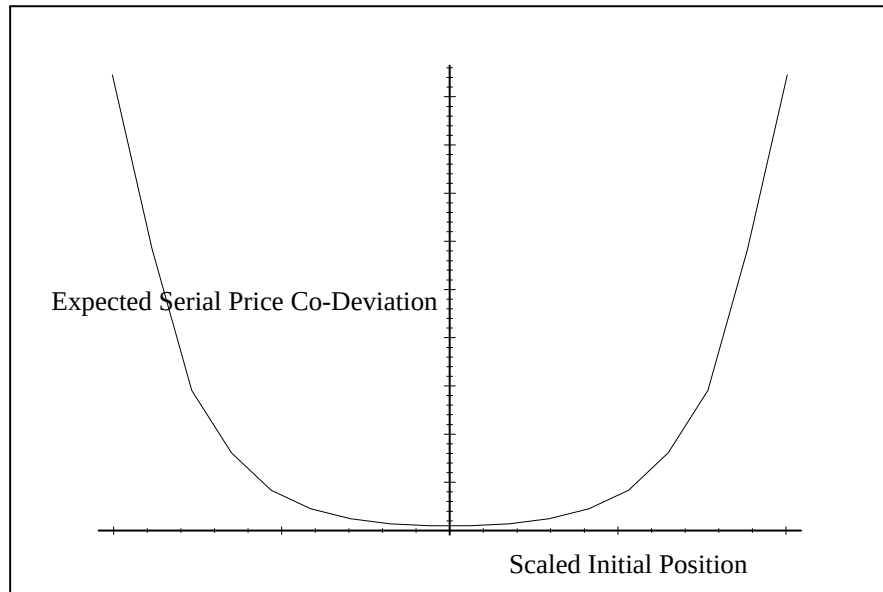


Figure 4: **Expected Serial Co-Deviation** as a function of the initial position. The horizontal axis is the initial position scaled by the maximum position that can be held by an arbitrageur who is willing to bear a 1% probability of bankruptcy in a given period. The following parameter values were used: $\sigma_F = 0.15$, $\sigma_I = 0.1$, $\beta = 1.0$ and the figure was generated using the exact numerical solution. A figure, not reported, was also generated using the approximation and was similar in shape and magnitude.

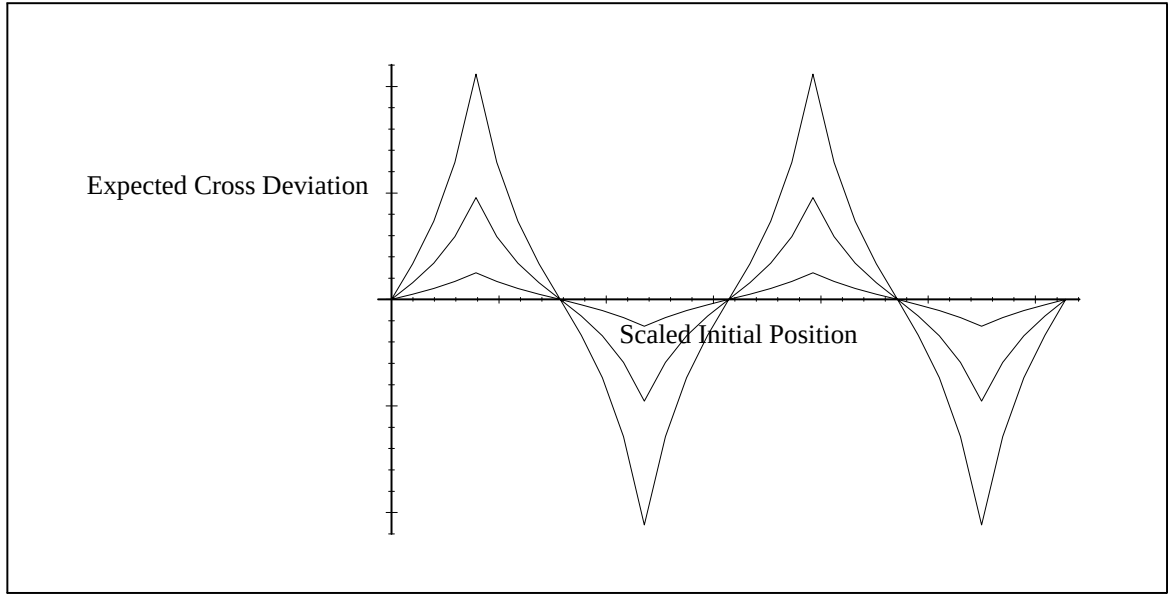


Figure 5: **Expected Cross Deviation** as a function of the arbitrageur's position in the two assets. The figure uses $\rho = 0$. The horizontal axis is $\tan^{-1} \left(\frac{\theta_{2,t-1}}{\theta_{1,t-1}} \right)$ over the range $(0, 2\pi)$. The three lines in the figure are for three different levels of $\sqrt[2]{\theta_{1,t-1}^2 + \theta_{2,t-1}^2}$, corresponding to 30%, 60% and 90% of maximum sustainable position. The following parameter values were used: $\sigma_F = 0.15$, $\sigma_I = 0.1$, $\beta = 1.0$ and the figure was generated using the approximate solution.

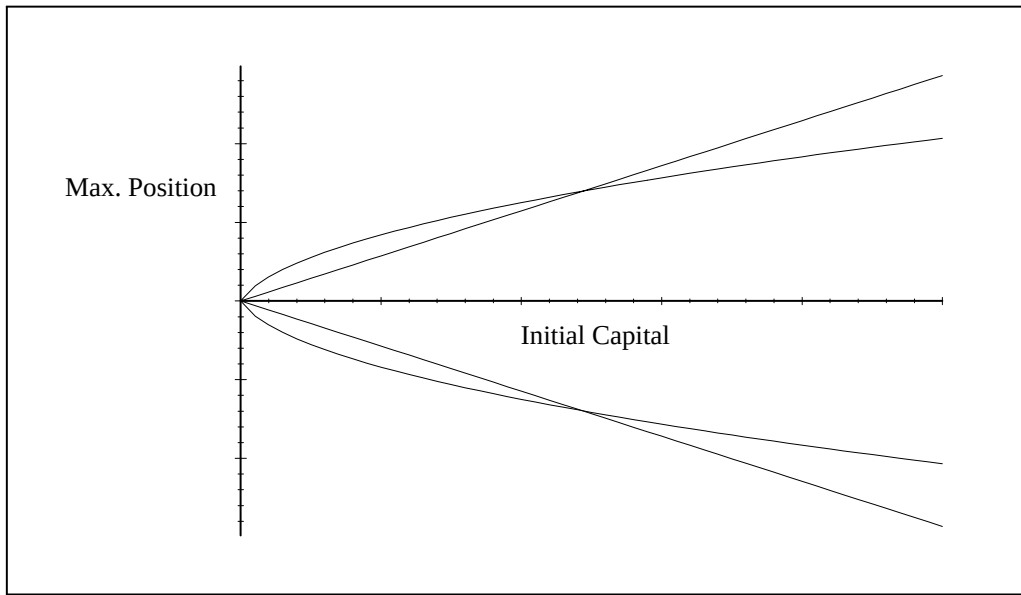


Figure 6: **Maximum sustainable position** as a function of the capital balance (EC_t). The curved lines are the values of θ_t that solve (30) with the LHS set to zero for a given level of probability. The straight lines correspond to the maximum sustainable position for a given margin requirement.

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